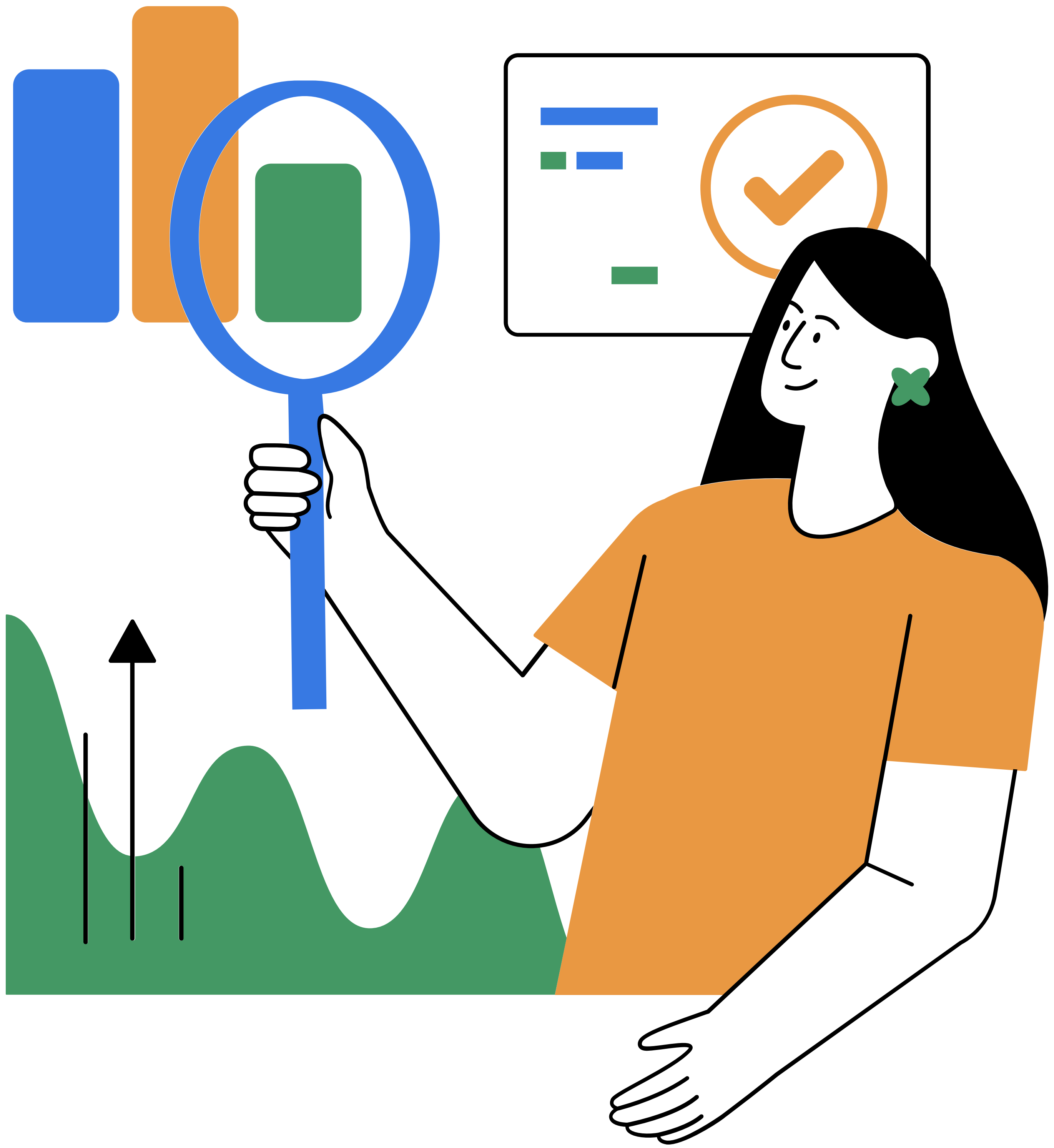
Data Visualisation - Customer Behavior in Telecom



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Introduction

Dataset Overview

- **Dataset:** Telco Customer Churn (from Kaggle)
- **Customers**: 7,043
- **Columns:** 21
- **Source**: https://www.kaggle.com/datasets/blastchar/telco-customer-churn
- **Info includes:**

1. **Customer info:** gender, senior citizen, partner, dependents
2. **Services:** phone, internet, streaming, security
3. **Payments:** contract, billing method, monthly & total charges
4. **Churn:** Did the customer leave the company?

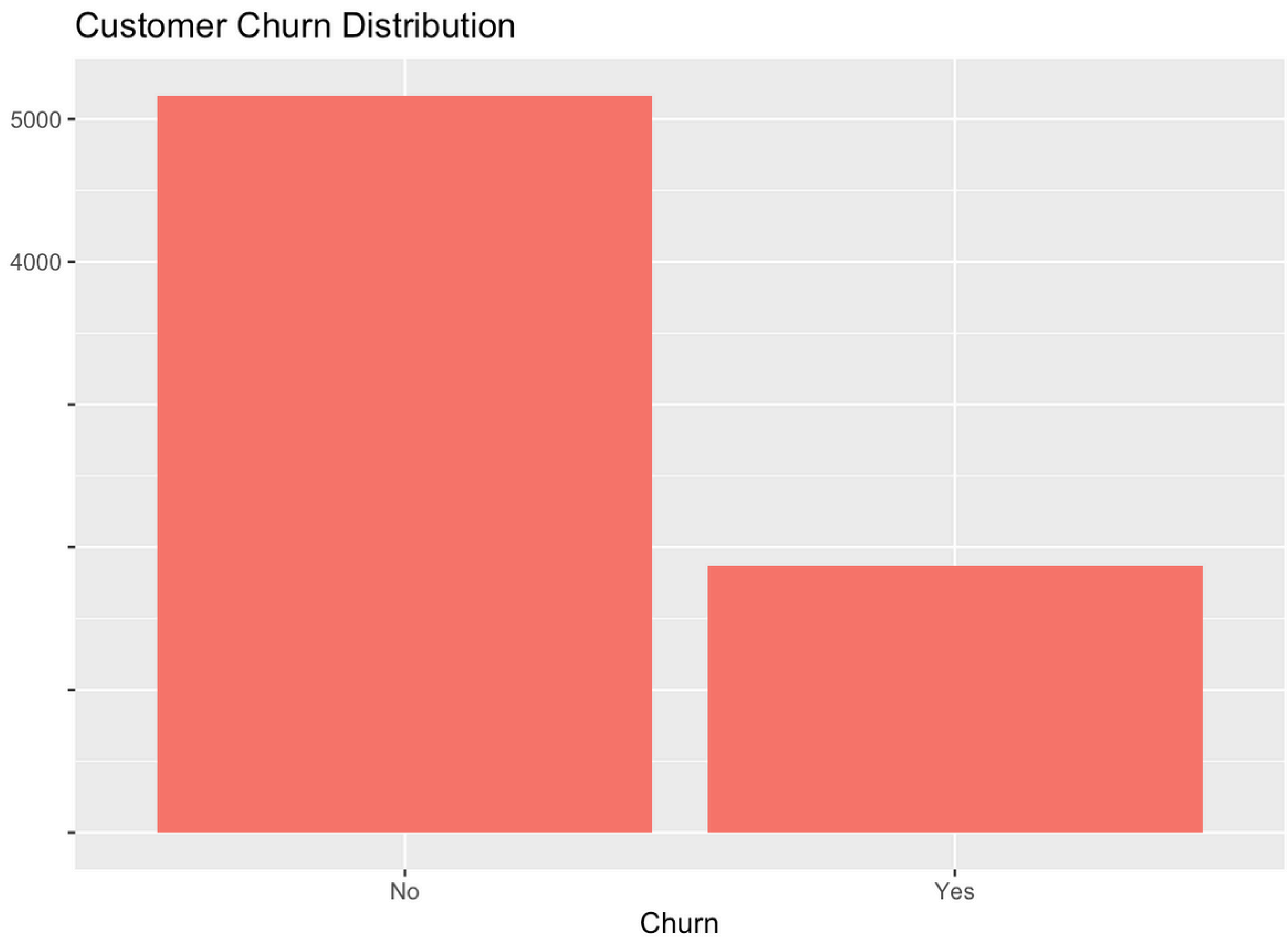
customerID	gender	SeniorCitizer	Partner	Dependents	tenure	PhoneService	MultipleLine	InternetServi
7590-VHVEG	Female	No	Yes	No	1	No	No phone se	DSL
5575-GNVDE	Male	No	No	No	34	Yes	No	DSL
3668-QPYBK	Male	No	No	No	2	Yes	No	DSL
7795-CFOCW	Male	No	No	No	45	No	No phone se	DSL
9237-HQITU	Female	No	No	No	2	Yes	No	Fiber optic
9305-CDSKC	Female	No	No	No	8	Yes	Yes	Fiber optic
1452-KIOVK	Male	No	No	Yes	22	Yes	Yes	Fiber optic
6713-OKOM	Female	No	No	No	10	No	No phone se	DSL
7892-POOKP	Female	No	Yes	No	28	Yes	Yes	Fiber optic
6388-TABGU	Male	No	No	Yes	62	Yes	No	DSL
9763-GRSKD	Male	No	Yes	Yes	13	Yes	No	DSL
7469-LKBCI	Male	No	No	No	16	Yes	No	No
8091-TTVAX	Male	No	Yes	No	58	Yes	Yes	Fiber optic
0280-XJGEX	Male	No	No	No	49	Yes	Yes	Fiber optic
5129-JLPIS	Male	No	No	No	25	Yes	No	Fiber optic
3655-SNQYZ	Female	No	Yes	Yes	69	Yes	Yes	Fiber optic
8191-XWSZG	Female	No	No	No	52	Yes	No	No
9959-WOFKT	Male	No	No	Yes	71	Yes	Yes	Fiber optic
4190-MFLUW	Female	No	Yes	Yes	10	Yes	No	DSL
4183-MYFRB	Female	No	No	No	21	Yes	No	Fiber optic
8779-QRDMV	Male	Yes	No	No	1	No	No phone se	DSL
1680-VDCWW	Male	No	Yes	No	12	Yes	No	No
1066-JKSGK	Male	No	No	No	1	Yes	No	No
3638-WEABW	Female	No	Yes	No	58	Yes	Yes	DSL
6322-HRPFA	Male	No	Yes	Yes	49	Yes	No	DSL
6865-JZNKO	Female	No	No	No	30	Yes	No	DSL

Data Pre-Processing Steps

- Handle Missing Values in TotalCharges by Removing 11 records
- Count the number of unique values in each variable to understand data diversity and potential categorical variables.
- Ensure there are no empty string values (" ") in the TotalCharges column.
- Transform numeric 0/1 encoding into categorical "No"/"Yes" for better interpretability in analysis and visualization.

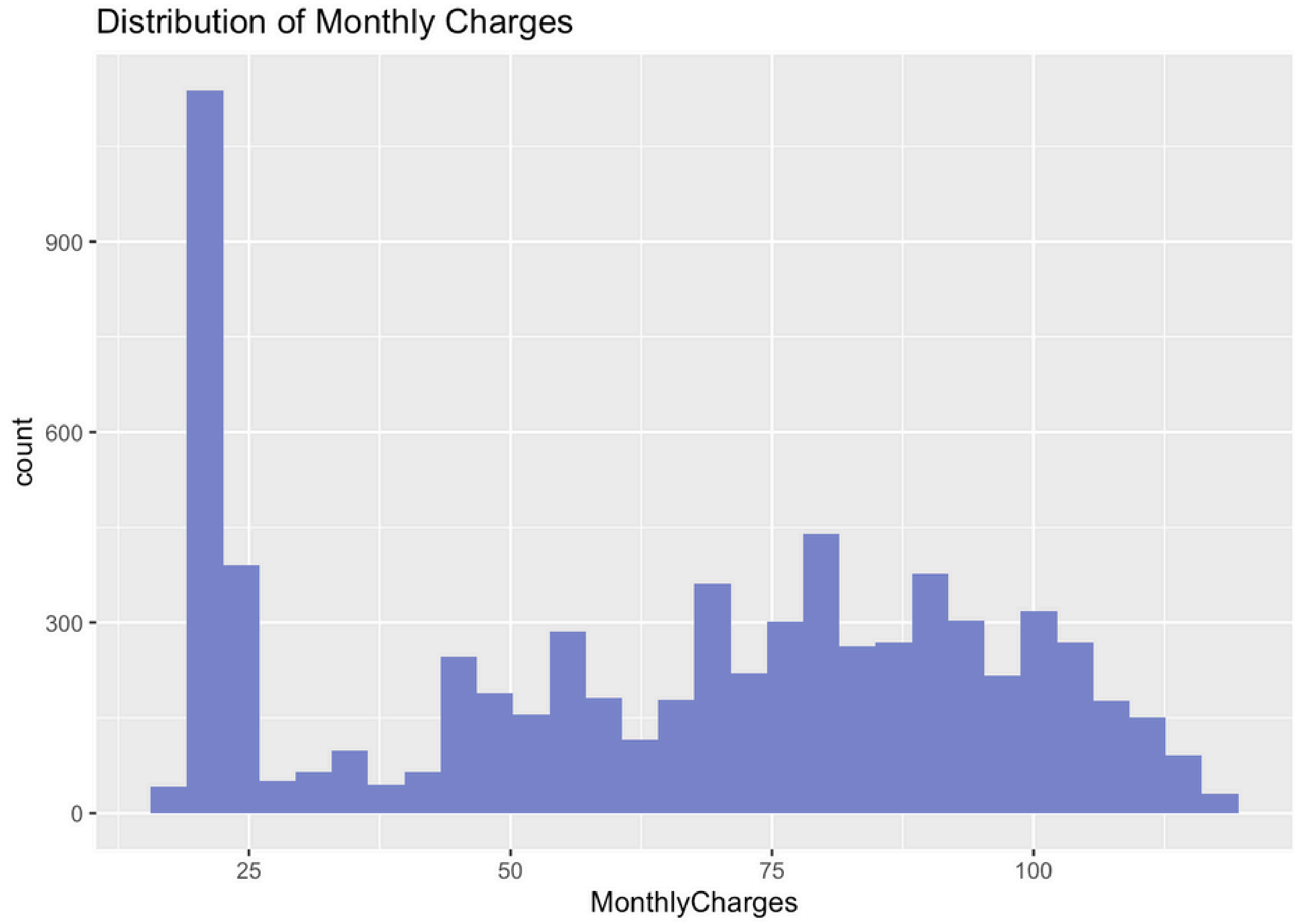
Bar Plots – Categorical Variables

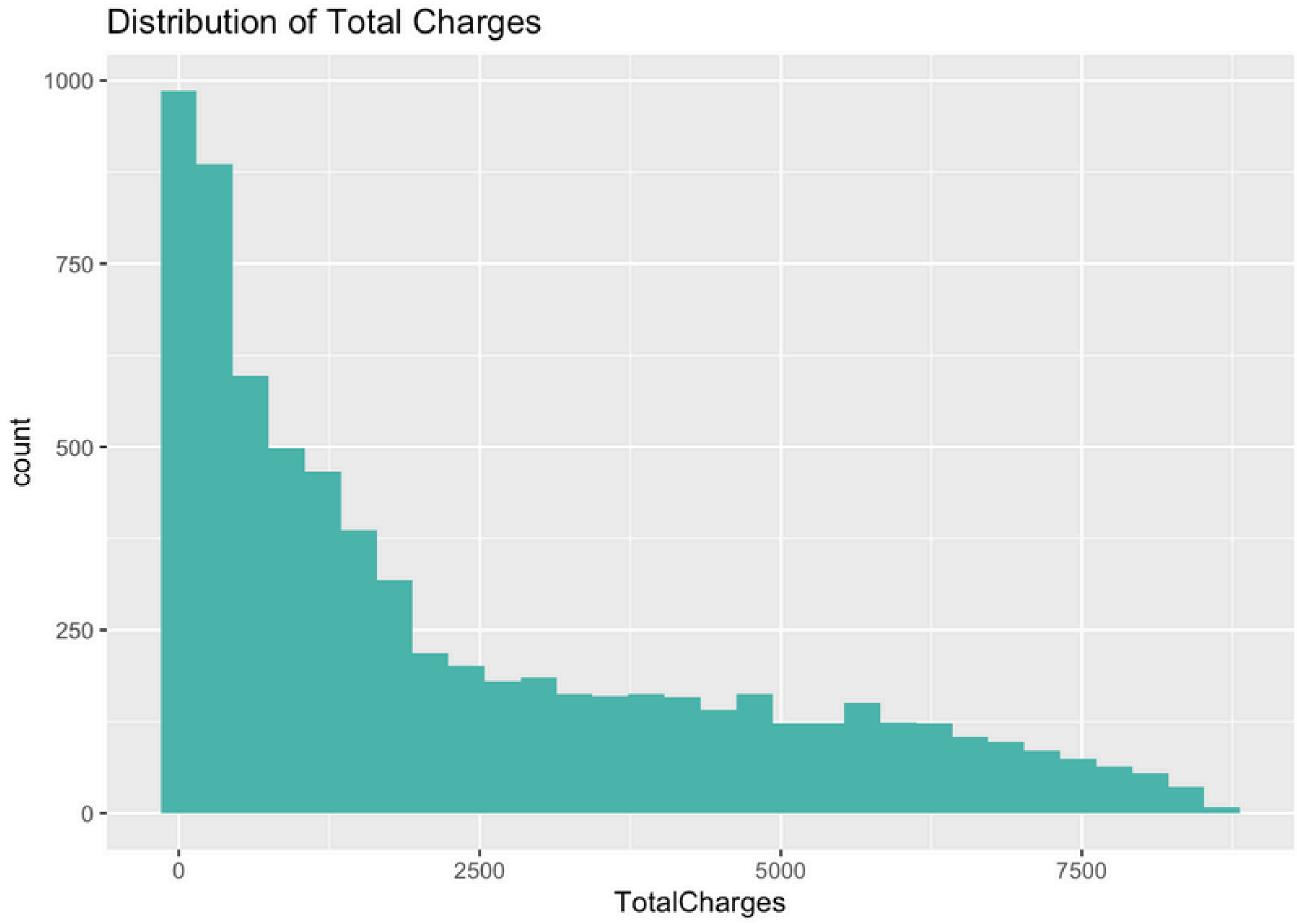
- Show distribution of customer categories.
- Most customers stay, fewer churn.
- Other categories: gender, senior citizen, partner, dependents, etc.

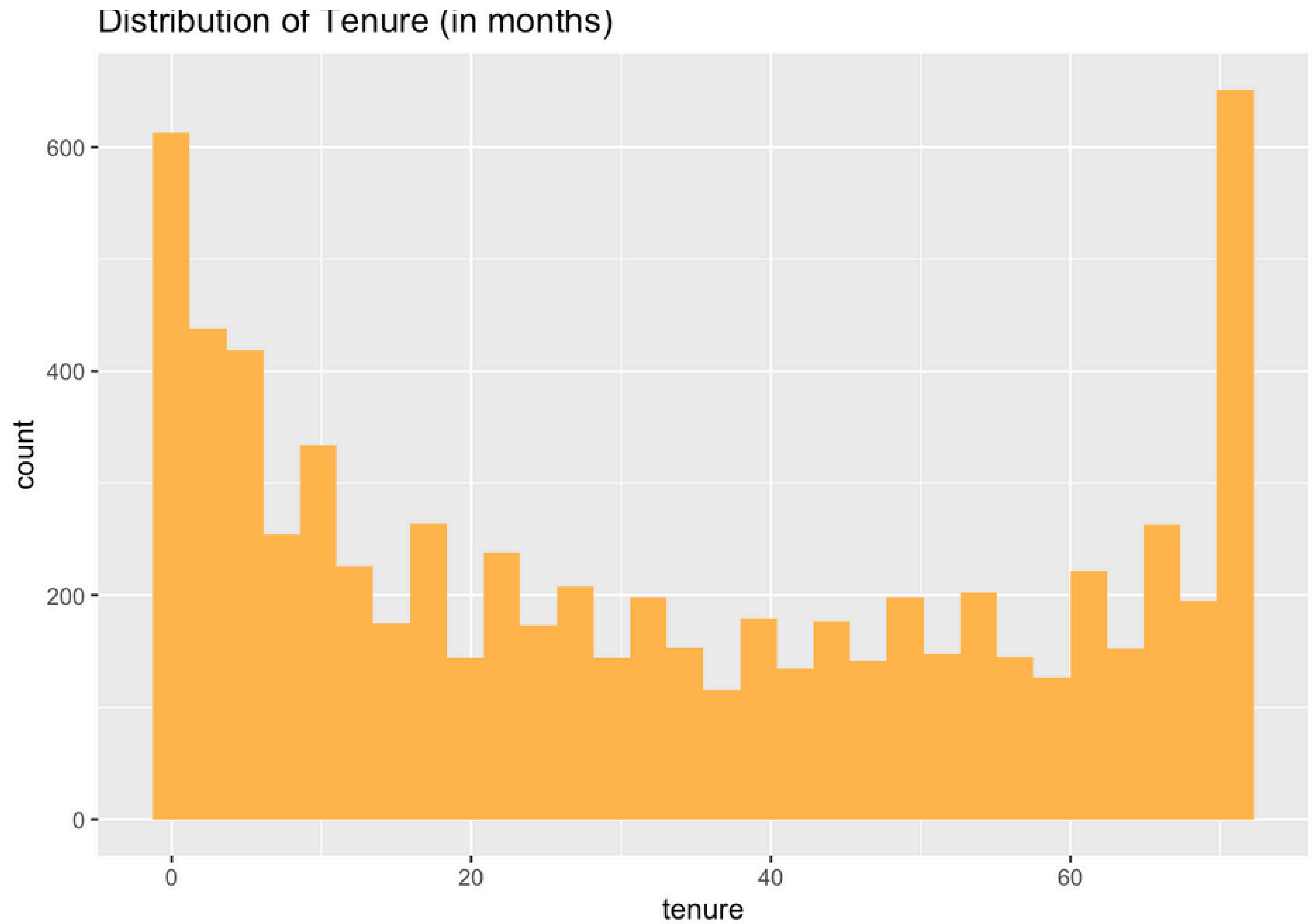


Histograms – Numeric Variables

- Tenure: U-shaped — many new and many loyal customers.
- MonthlyCharges: two peaks — low and high bill groups.
- TotalCharges: right-skewed — many new customers, few with high totals.

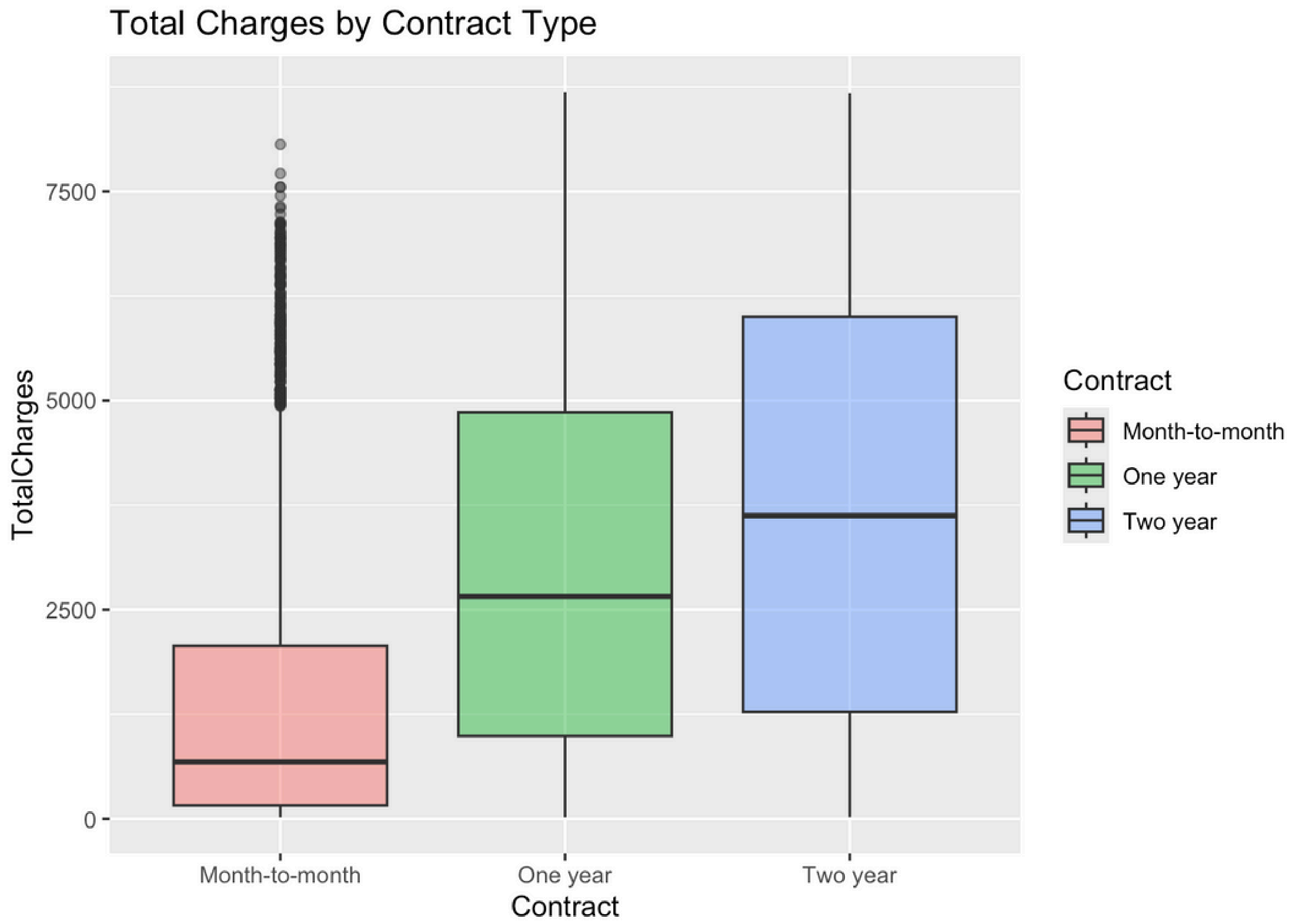


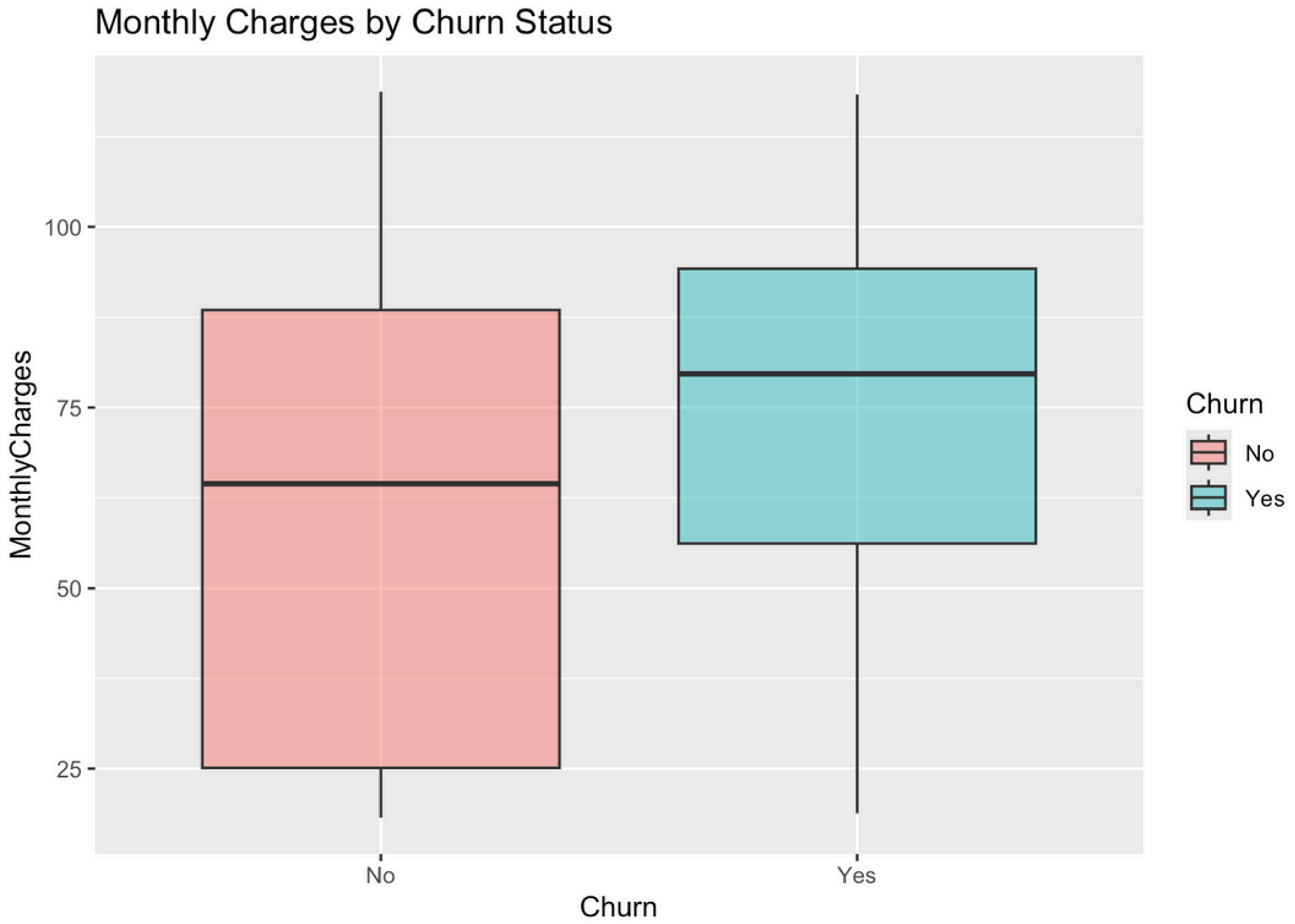




Box Plots – Spotting Outliers

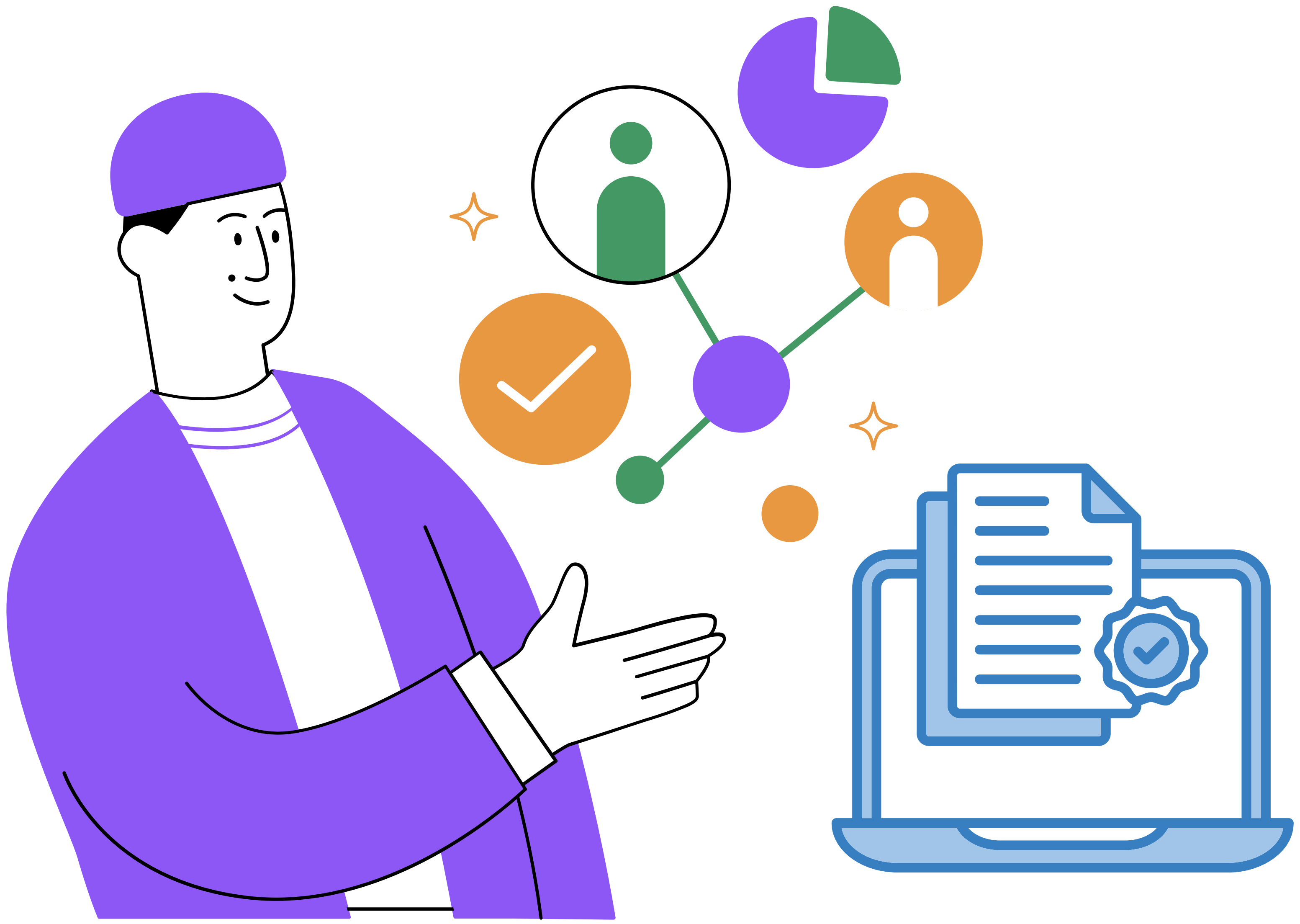
- Monthly charges: churned customers pay more.
- Total charges: increase with longer contracts.
- Some month-to-month customers have very high totals.





Question 1

How Do Customer Demographics and Contract Types Influence Spending and Churn?

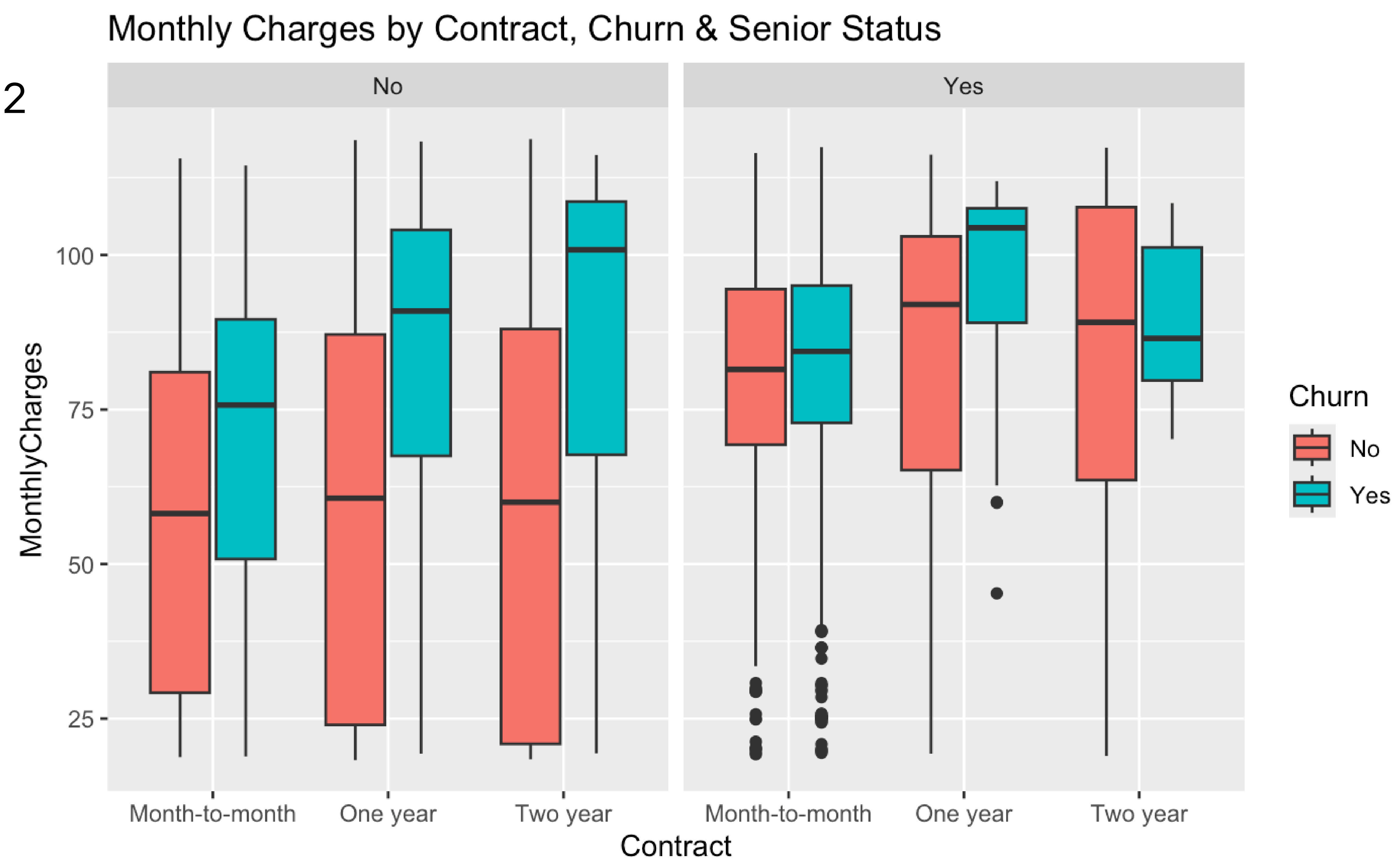


Objective of the Analysis

- **Which groups spend more money?**
  - Men or women? Seniors or younger people? With kids or not?
- **Does contract type change spending?**
  - Do people with long contracts (like 1 year or 2 years) pay less?
- **Do people who leave pay more or less?**
  - Is spending linked to churn?

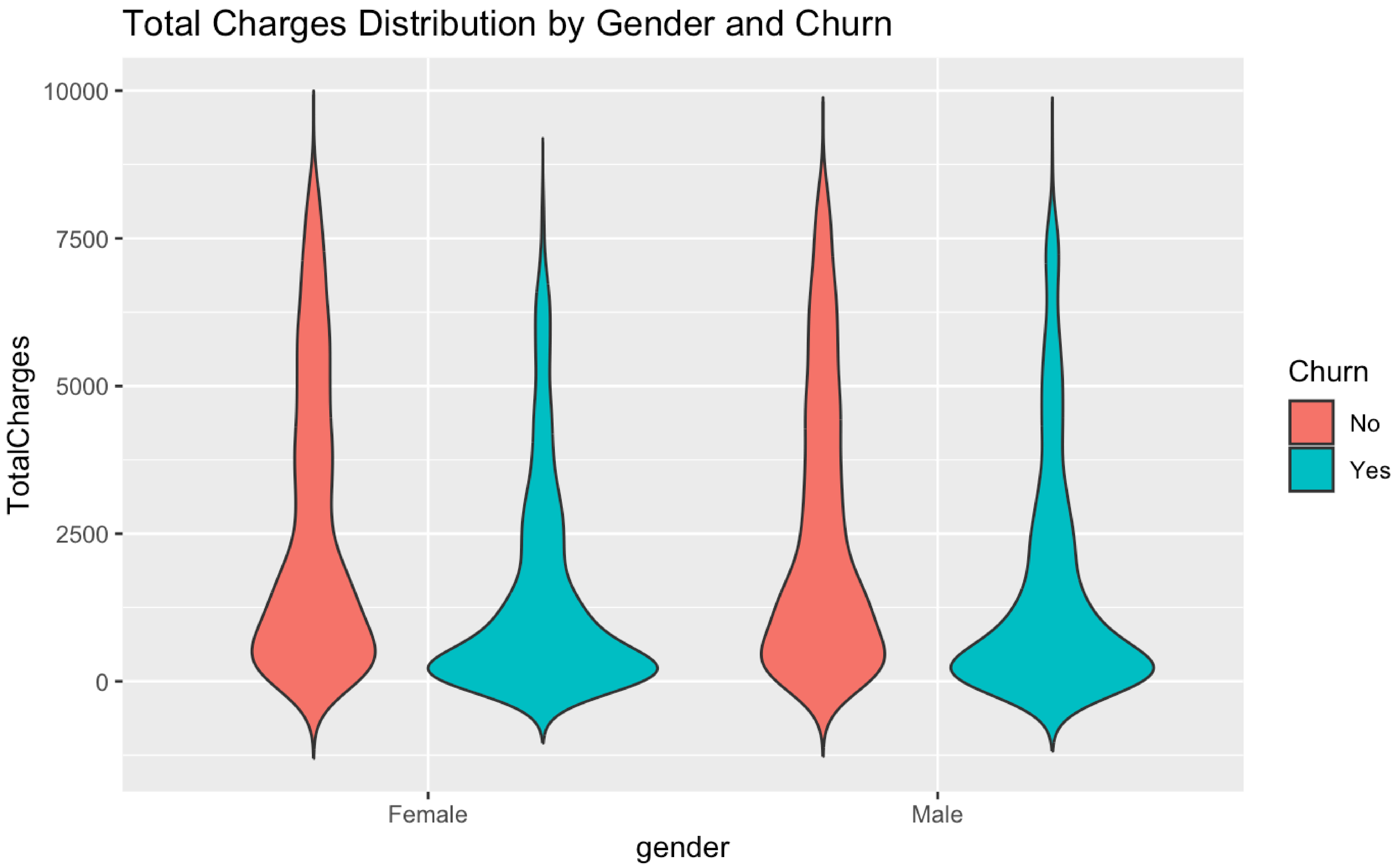
Monthly Charges by Contract Type, Churn, and Senior

- **X-axis:** Contract type (Month-to-month, 1 year, 2 years)
- **Colors:** Churn (Yes / No)
- **Facets**: Senior Citizen (Yes / No)
- **Key findings**:
- Month-to-month → higher and more spread charges.
- Churn = Yes customers pay more monthly.
- Senior citizens usually spend a bit less.



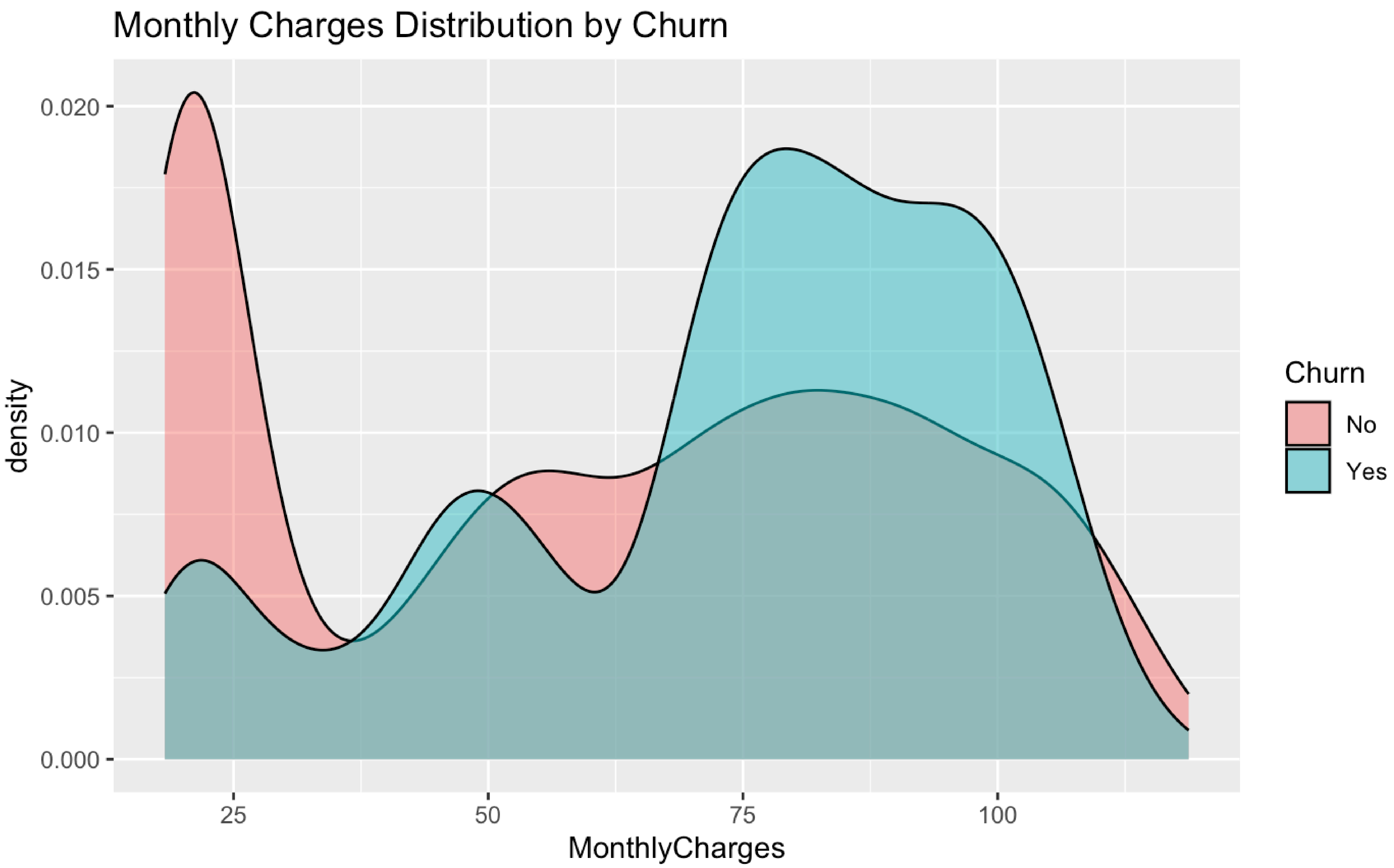
Total Spending by Gender and Churn

- Red = people who stayed
- Blue = people who left
- People who stayed spent more in total
- Gender doesn't really change the pattern



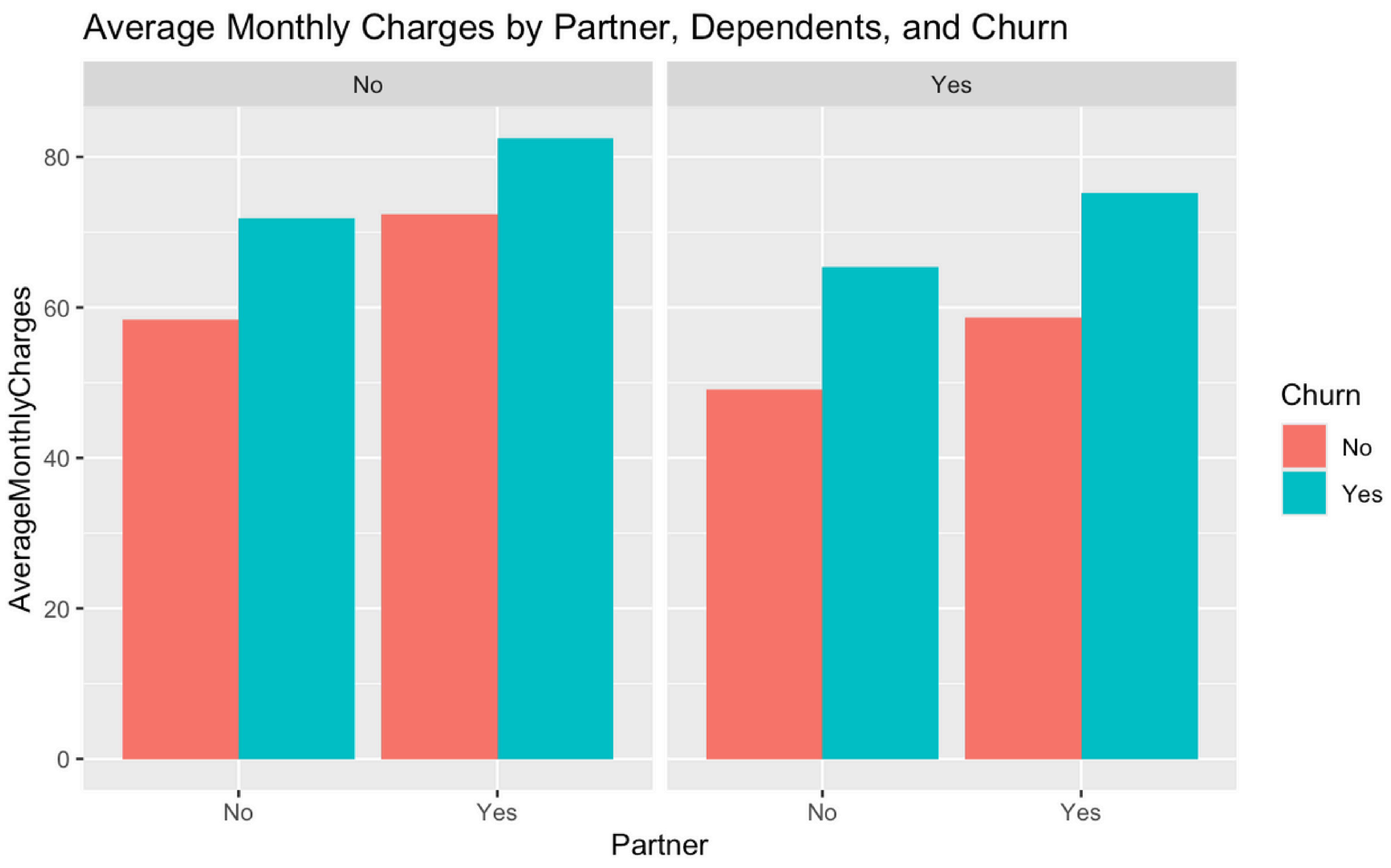
Density Plot – Monthly Charges by Churn

- Shows spread of monthly charges for Churn = Yes vs No.
- Churn = Yes curve is wider and shifted right → higher monthly bills.
- Churn = No curve is left and tighter → lower, stable bills.



Grouped Bar Chart – Average Monthly Charges by Partner, Dependents & Churn

- **Two panels:** Dependents = No (left), Dependents = Yes (right).
- **X-axis:** Partner (No / Yes).
- **Colors:** Churn (Yes / No).
- **Key findings:**
- Churn = Yes customers pay more in every group.
- Having dependents → lower monthly spending.
- Highest charges: Partner = Yes, Dependents = No, Churn = Yes.

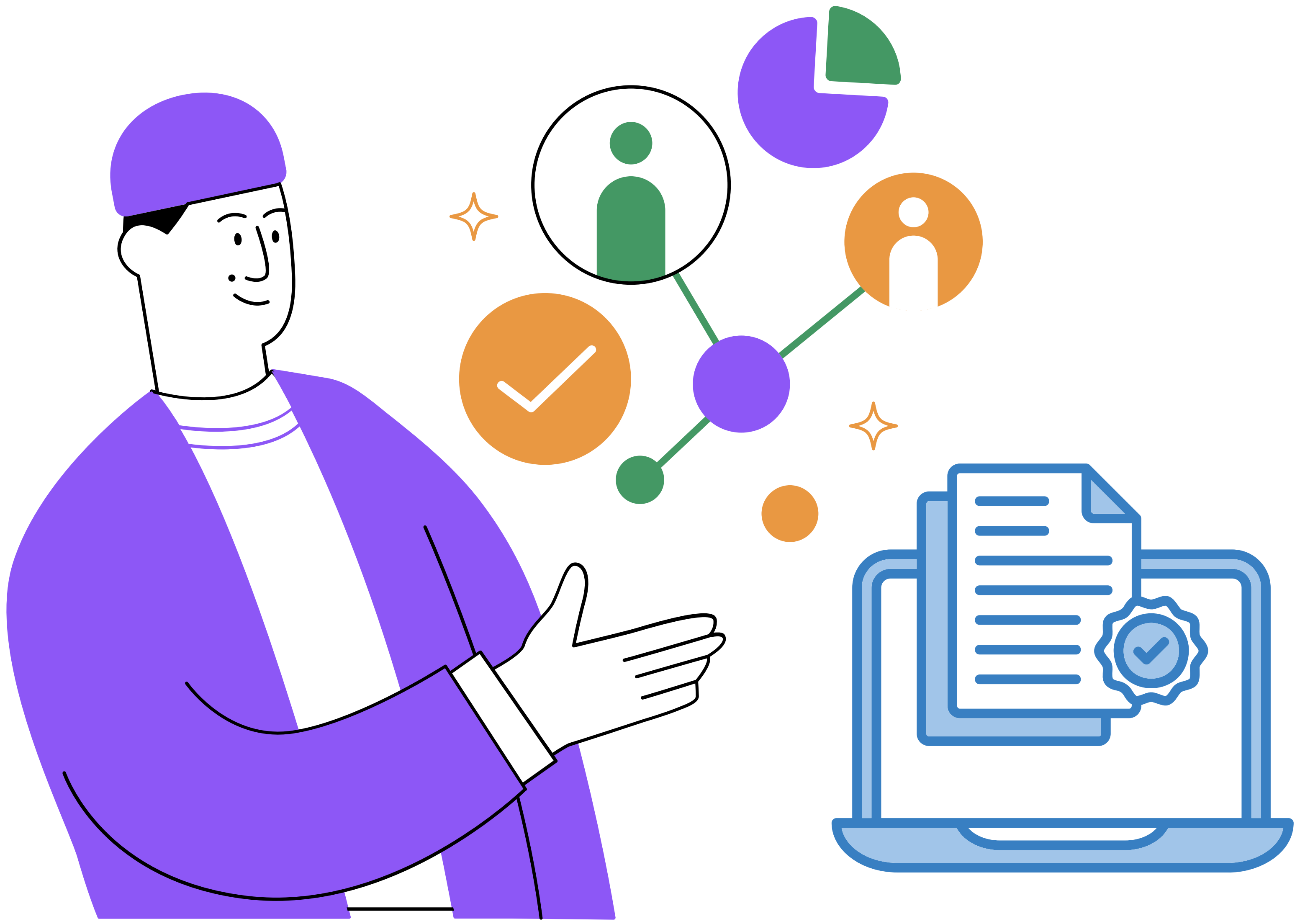


Key Business Takeaways

- **Pricing Strategy:** Higher churn among customers paying more suggests a need to reassess pricing or enhance perceived value to reduce cancellations.
- **Customer Segmentation**: Different spending patterns by family status highlight the importance of segmenting customers to tailor plans and offers effectively.
- **Targeted Retention**: Identifying high-risk groups (e.g., partnered, no dependents, high spenders) allows for focused retention campaigns that can protect revenue.
- **Product Development**: Insights into spending behavior can guide the design of personalized bundles or loyalty programs that better meet customer needs and reduce churn.

Question 2

How does customer loyalty (tenure) vary across contract and internet service types, and how does this relate to churn?



Objective of the Analysis

- **Understand customer loyalty**

Measure how long customers (tenure) typically stay based on contract and internet service types

- **Identify churn patterns**

Examine which combinations of contract type and internet service are associated with higher churn rates

- **Spot high-risk segments**

Detect customer groups that show early churn behavior

- **Support business decisions**

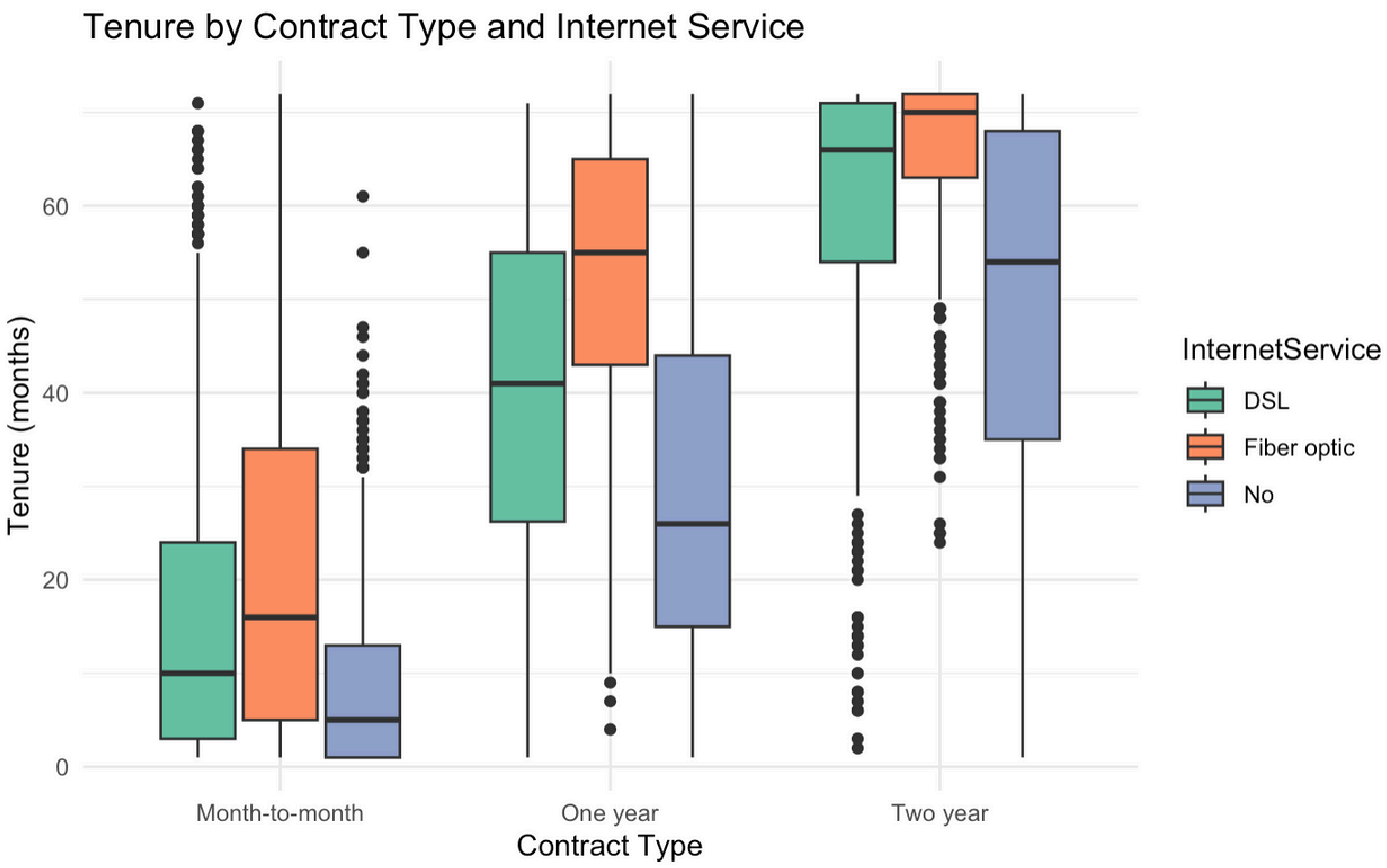
Provide data-backed insights to improve retention strategies and reduce customer churn

Box Plot - Tenure by Contract Type and Internet Services

- **Y-axis:** Tenure(Months)
- **X-axis**: Contract Type
- **Colors:** Internet Services

- **Key findings**:

1. Longer Contracts = Longer Tenure
2. Two-Year Contracts Show Highest Loyalty
3. Month-to-Month Contracts = Low Tenure

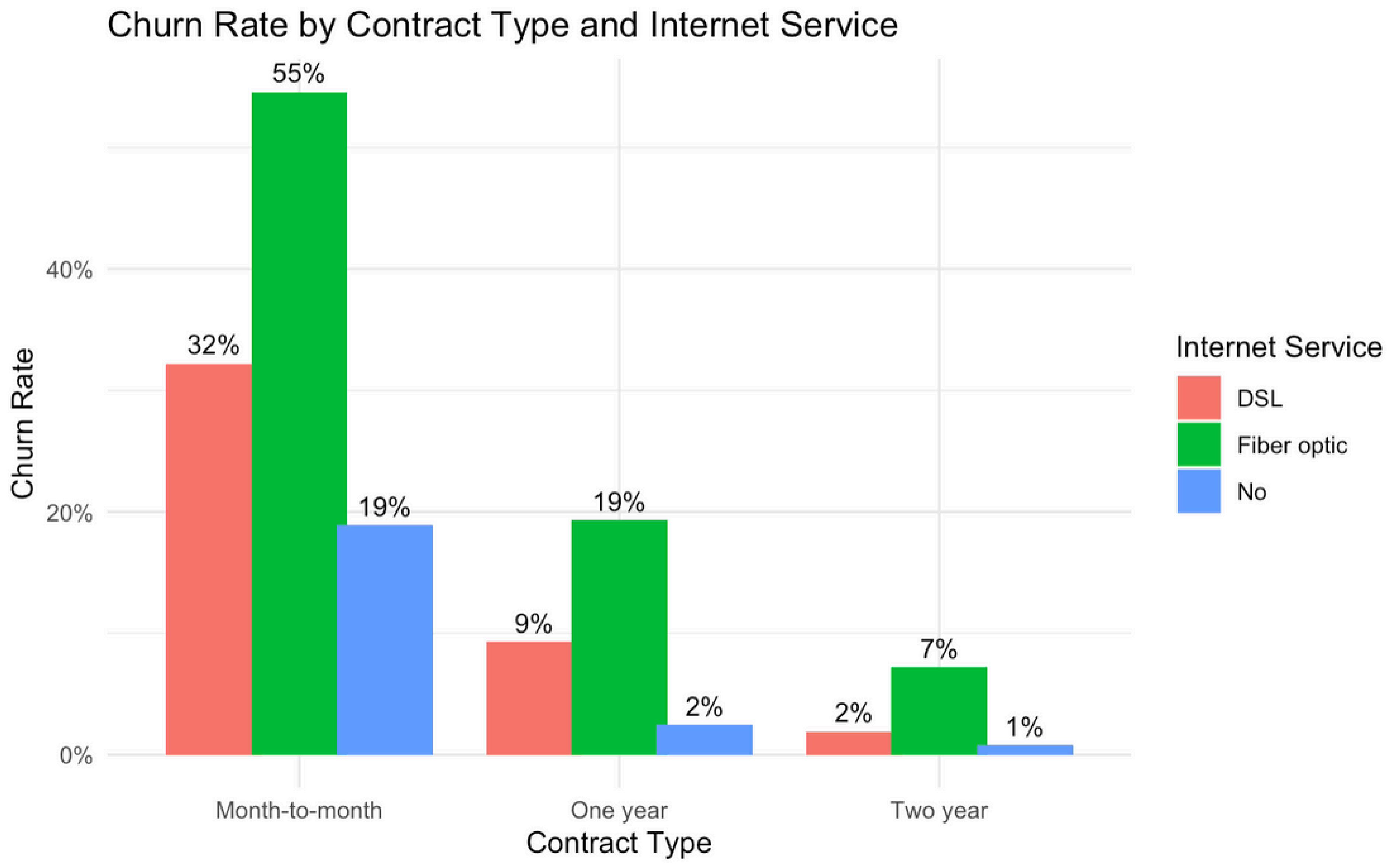


Bar Plot- Churn Rate by Contract Type and Internet Service

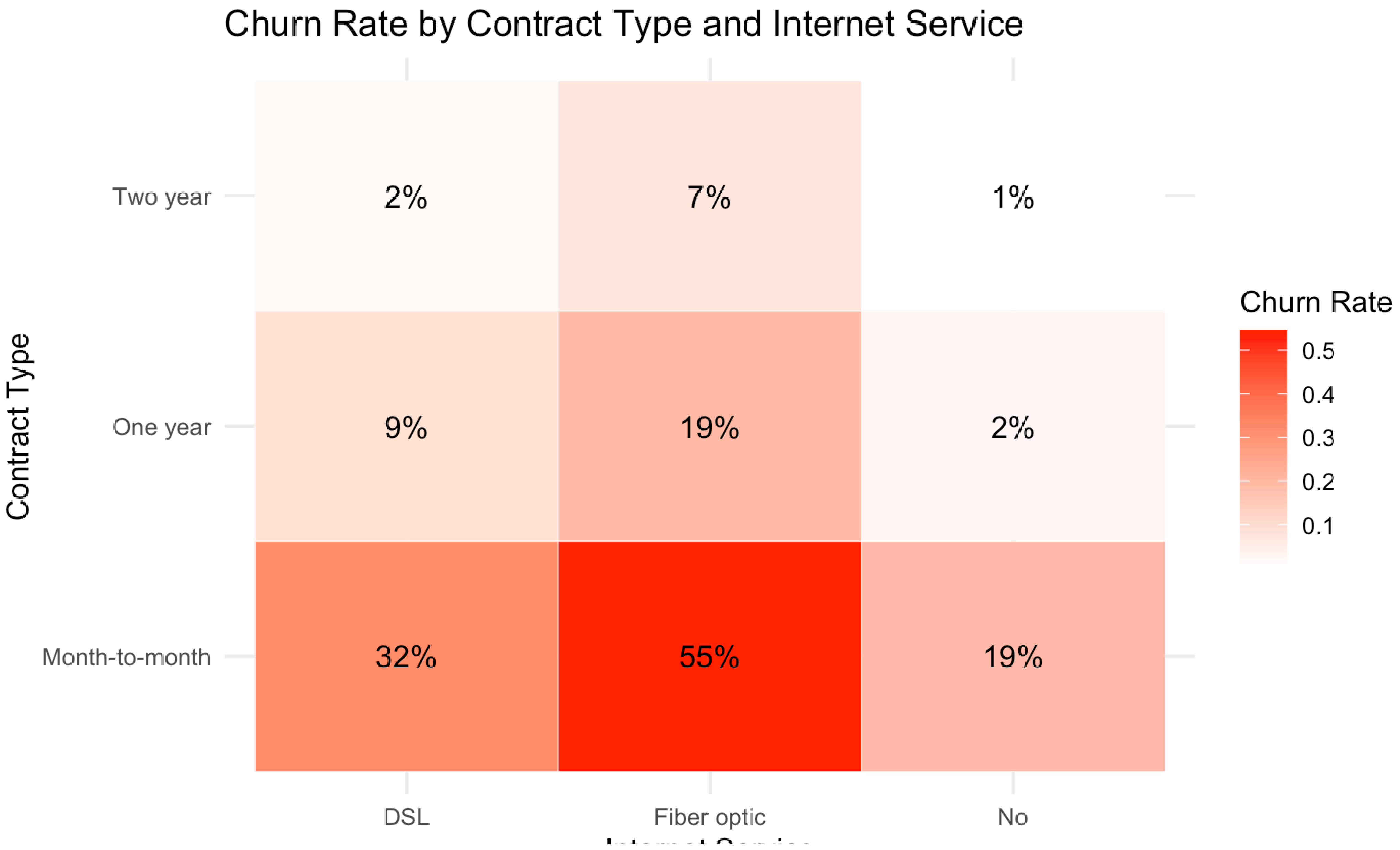
- **Y-axis:** Churn Rate
- **X-axis**: Contract Type
- **Colors:** Internet Services

- **Key findings**:

1. Month-to-month contracts have significantly higher churn rates than one-year or two-year contracts.
2. Two-Year Contracts Have Lowest Churn across all internet service types.
3. Customers with Fiber Optic on a month-to-month contract have the highest churn rate at 55% .
4. DSL service tends to have lower churn rates compared to Fiber Optic

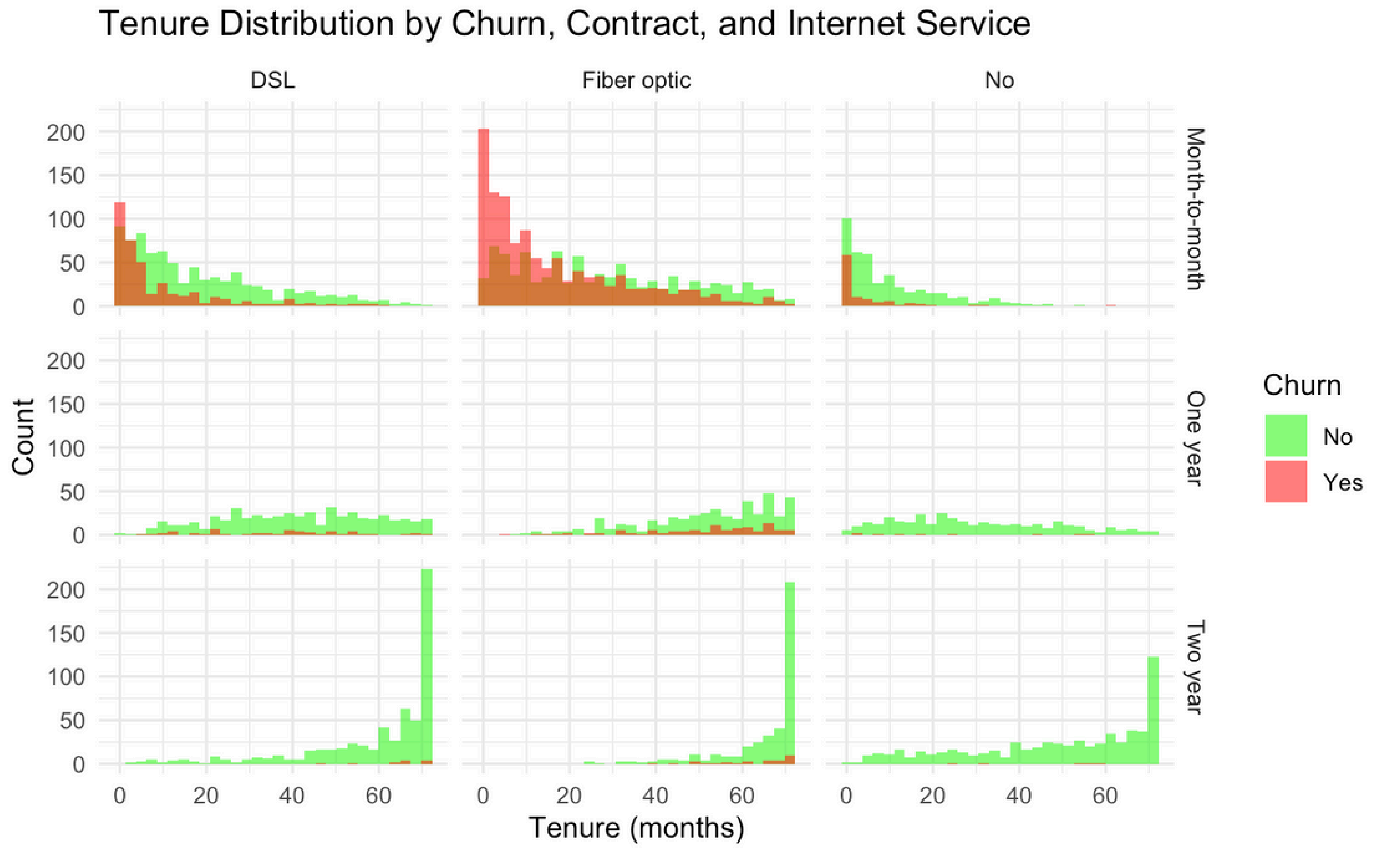


Heat Map


Churn Rate by Contract Type and Internet Service
Contract Type
Two year
One year
Month-to-month
2%
7%
1%
9%
19%
2%
32%
55%
19%
DSL
Fiber optic
No
Churn Rate
0.5
0.4
0.3
0.2
0.1

Histogram -Tenure Distribution by Churn, Contract, and Internet Service

- **Y-axis:** No. of Customers
- **X-axis**: Tenure (months)
- **Columns:** Internet Service (DSL, Fiber optic, No).
- **Rows:** Contract Type (Month-to-month, One year, Two year).
- **Color (within bars):** Churn Status (No - green, Yes - red).



- **Conclusion** :

1. Churners (red) primarily churn early
2. Non-churners (green) exhibit longer tenures
3. Longer contracts lead to longer retention
4. Month-to-month Fiber Optic is a high-risk group

Hypothesis Testing Framework

- **Conclusion** :
  1. **Tenure Effect** - longer a customer stays (higher tenure), the less likely they are to churn.
  2. **Contract Type Effect:** Customers with 2-year contracts are even less likely to churn — showing that long-term contracts promote loyalty.
  3. **Internet Service Effect**:

     Customers with Fiber Optic are more likely to churn than DSL users

     Customers with No internet are less likely to churn

```
Call:
glm(formula = Churn ~ tenure + Contract + InternetService, family = "binomial",
    data = data)

Coefficients:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                -0.30682    0.06294  -4.875 1.09e-06 ***
tenure                     -0.03139    0.00194 -16.180  < 2e-16 ***
ContractOne year           -0.84633    0.10216  -8.284  < 2e-16 ***
ContractTwo year           -1.67476    0.16880  -9.921  < 2e-16 ***
InternetServiceFiber optic  1.19854    0.07248  16.536  < 2e-16 ***
InternetServiceNo          -1.04584    0.11736  -8.911  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 8143.4  on 7031  degrees of freedom
Residual deviance: 6064.1  on 7026  degrees of freedom
AIC: 6076.1

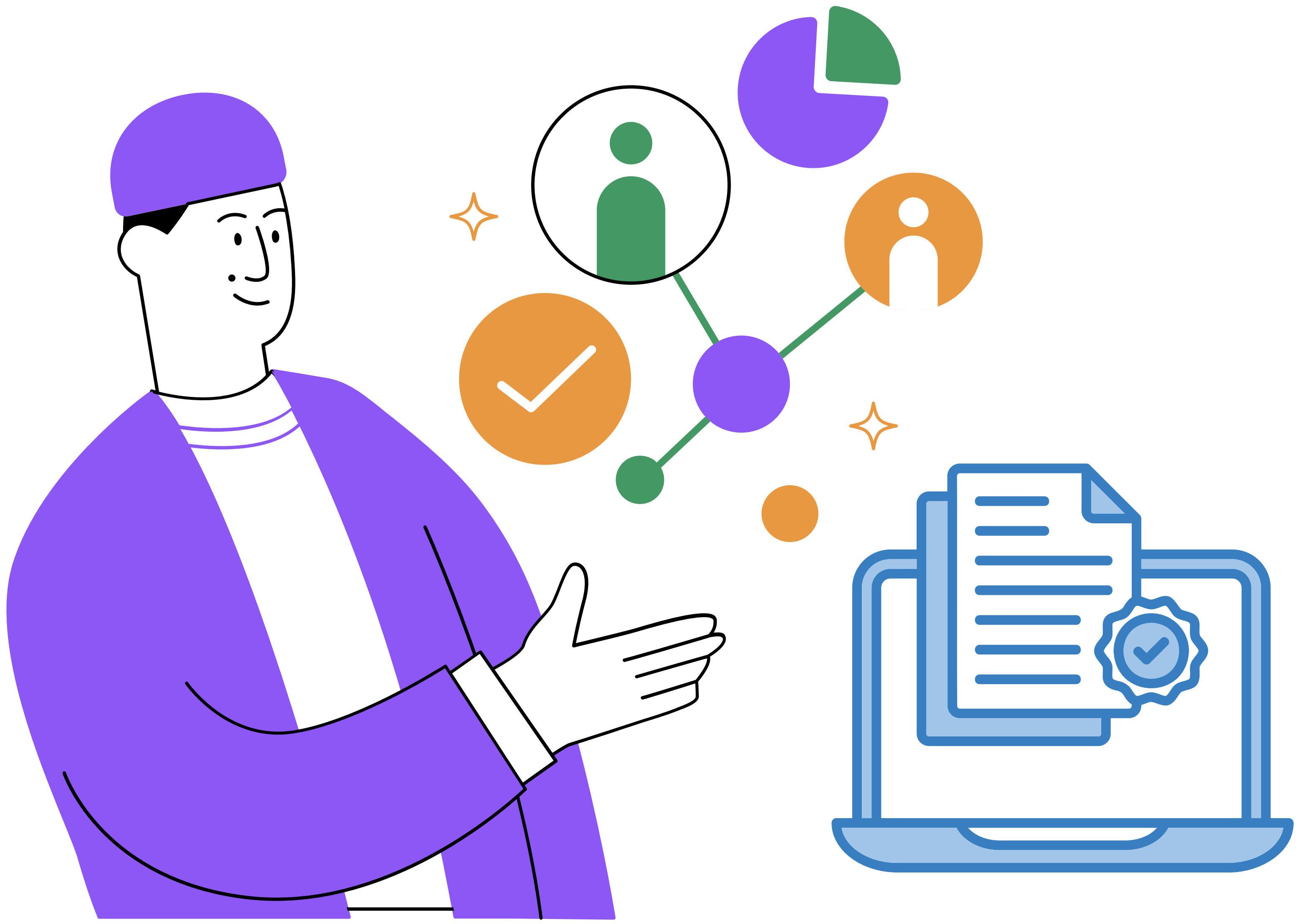
Number of Fisher Scoring iterations: 6
```

Key Business Takeaways

- Retention strategies should prioritize new customers and those in month-to-month fiber optic plans, especially during their initial months.
- Promoting longer contract terms may significantly enhance customer loyalty and reduce churn.
- Special attention and tailored offers or service improvements might be needed for fiber optic customers to reduce their higher churn propensity.
- Early-tenure churn is critical — addressing customer experience in the first few months can prevent most churn events

Question 3

Does having OnlineSecurity and TechSupport affect how long customers stay and how much they spend? And does this effect change depending on the type of contract they have?

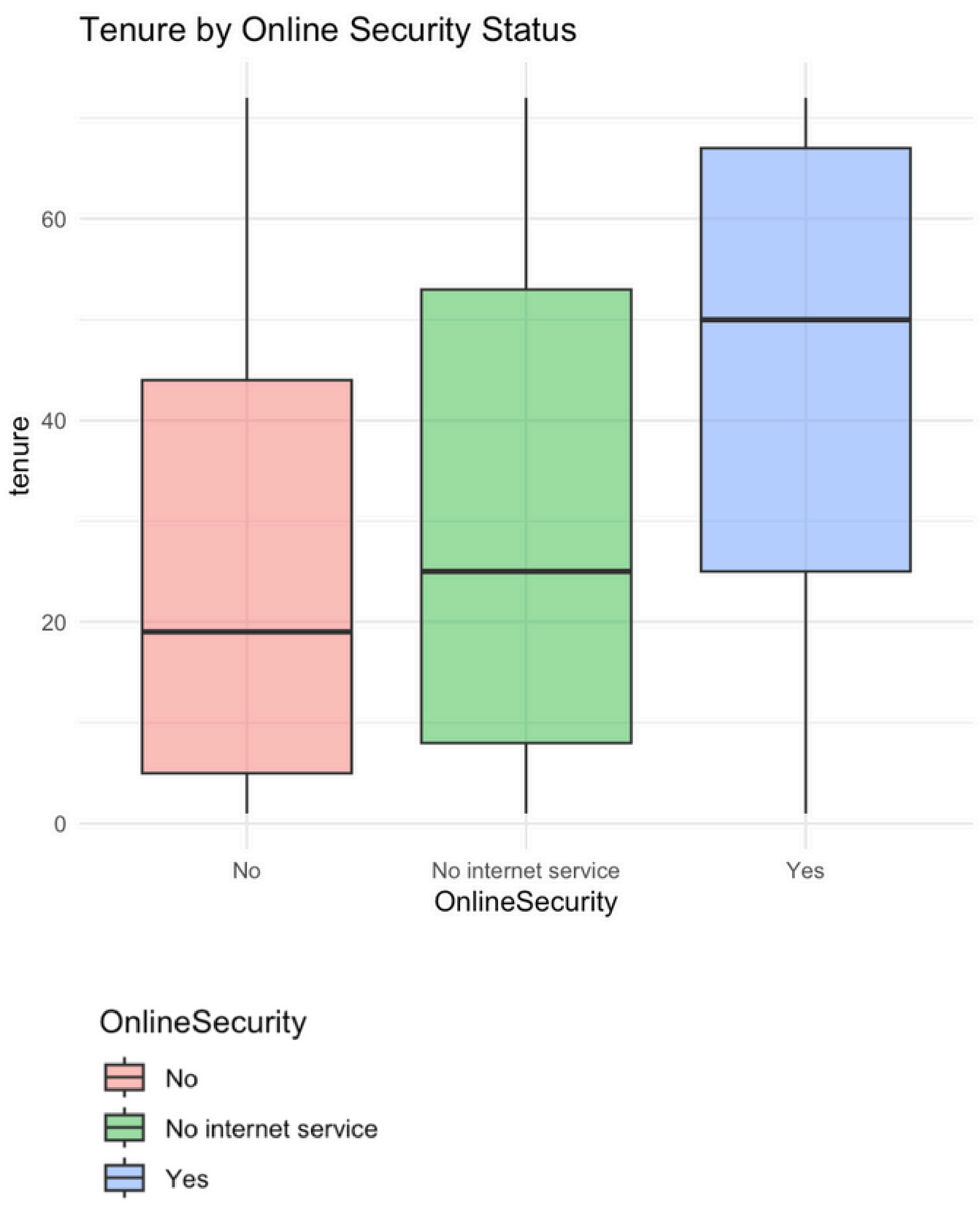


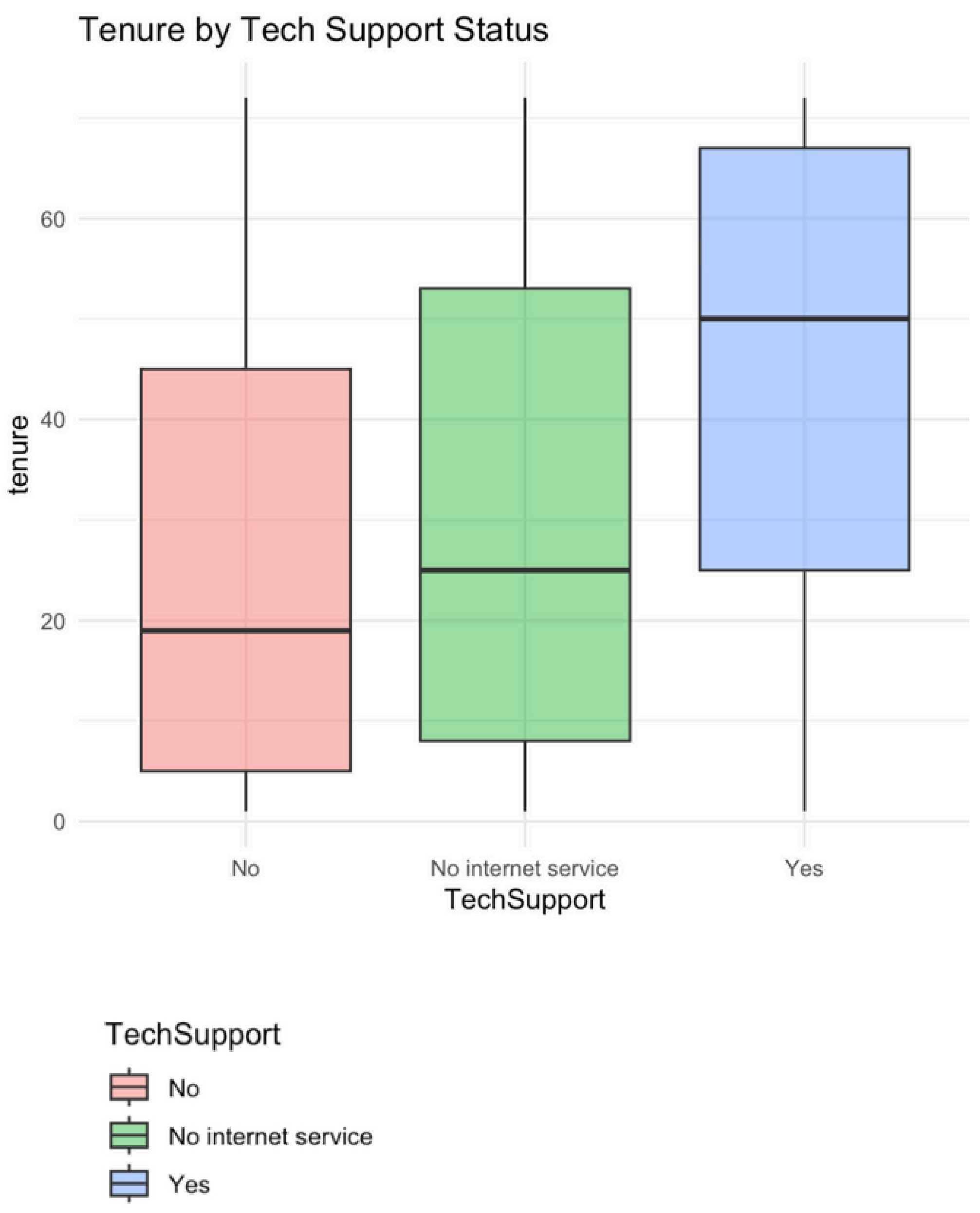
Objective of the Analysis

- Do **OnlineSecurity** and **TechSupport** services lead to longer customer retention?
- Do customers with these services **spend more** per month?
- Does the effect of these services **change by contract type** (e.g., month-to-month vs. yearly)?
- Does the impact of these services vary for different contract types?

Tenure by Online Security and Tech Support

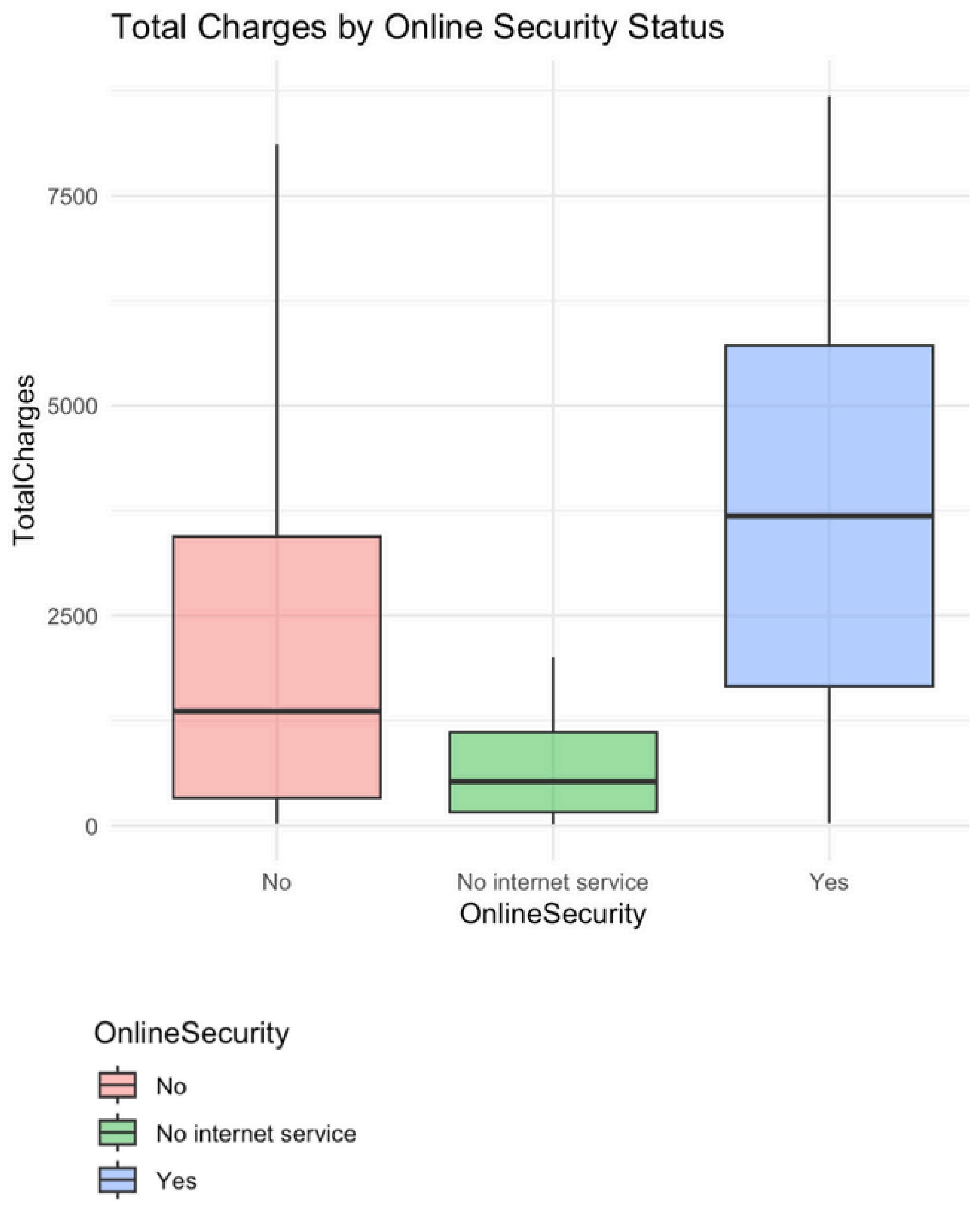
- **Key findings**:
    - Customers with Online Security service have the longest tenures.
    - Followed by no internet service - may have subscribed to a different service package (such as phone-only)
    - Least - ones that lack Online Security
    - Almost identical pattern for Tech support
    - Reinforces idea that lack of support services associated with earlier churn rates

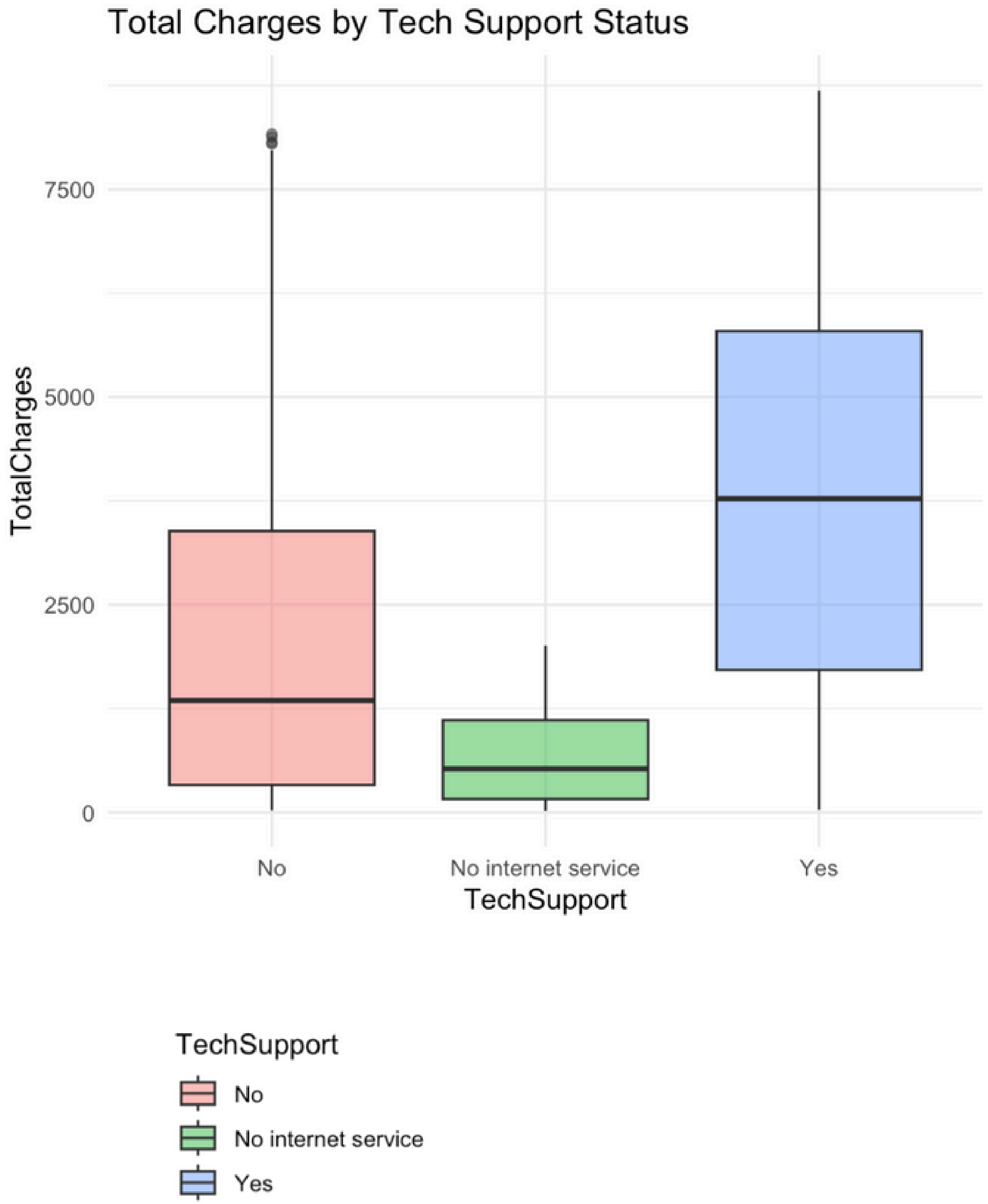




Total Charges by Online Security and Tech Support

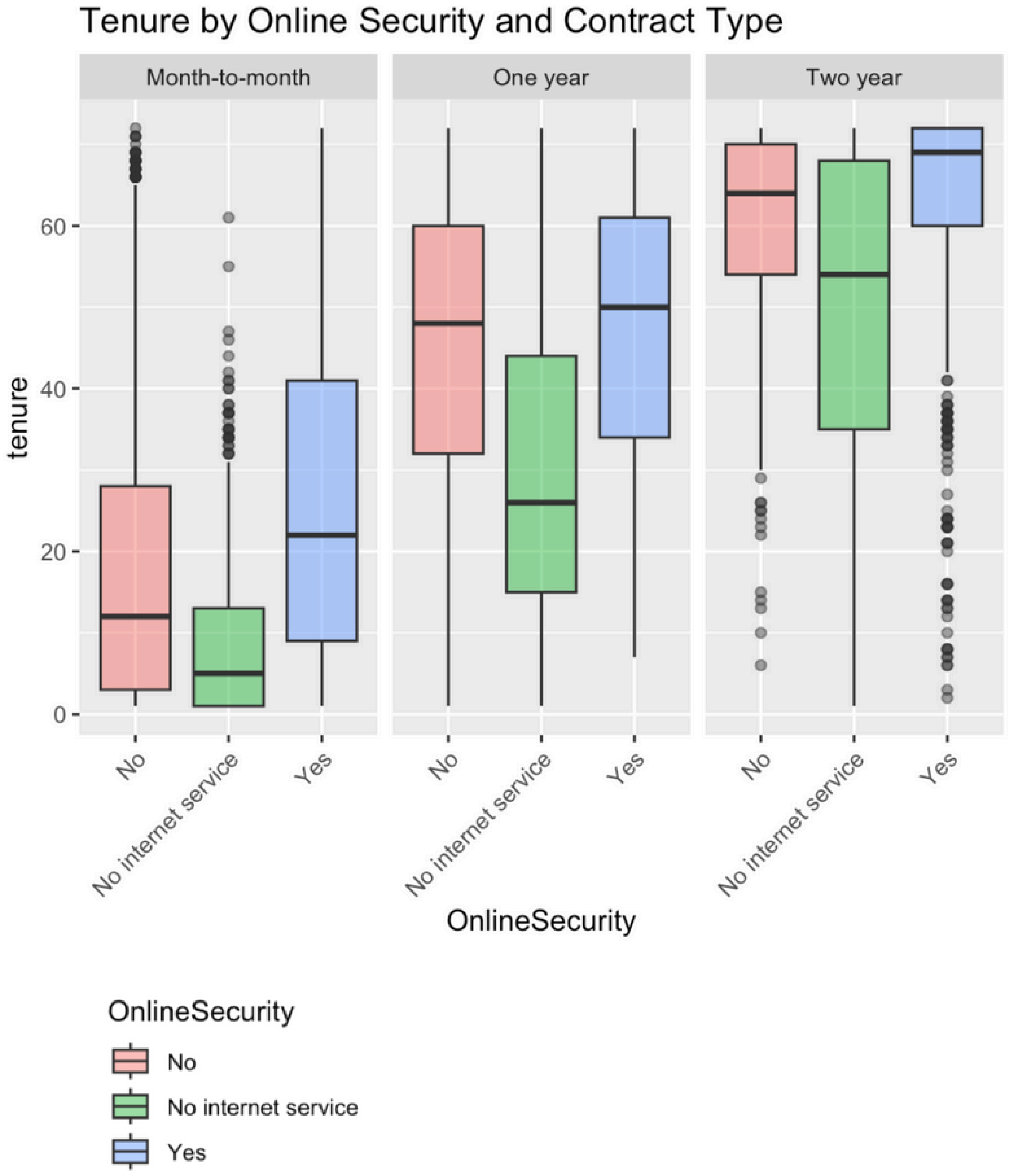
- **Key findings**:
  - Graphs for both services almost identical
  - Customers with both services have higher charges (as expected)
  - Those with no internet service have lower costs (no additional charges)
  - No service charges - have lower charges but more than no internet service customers (pay for internet, but not for extra services.)

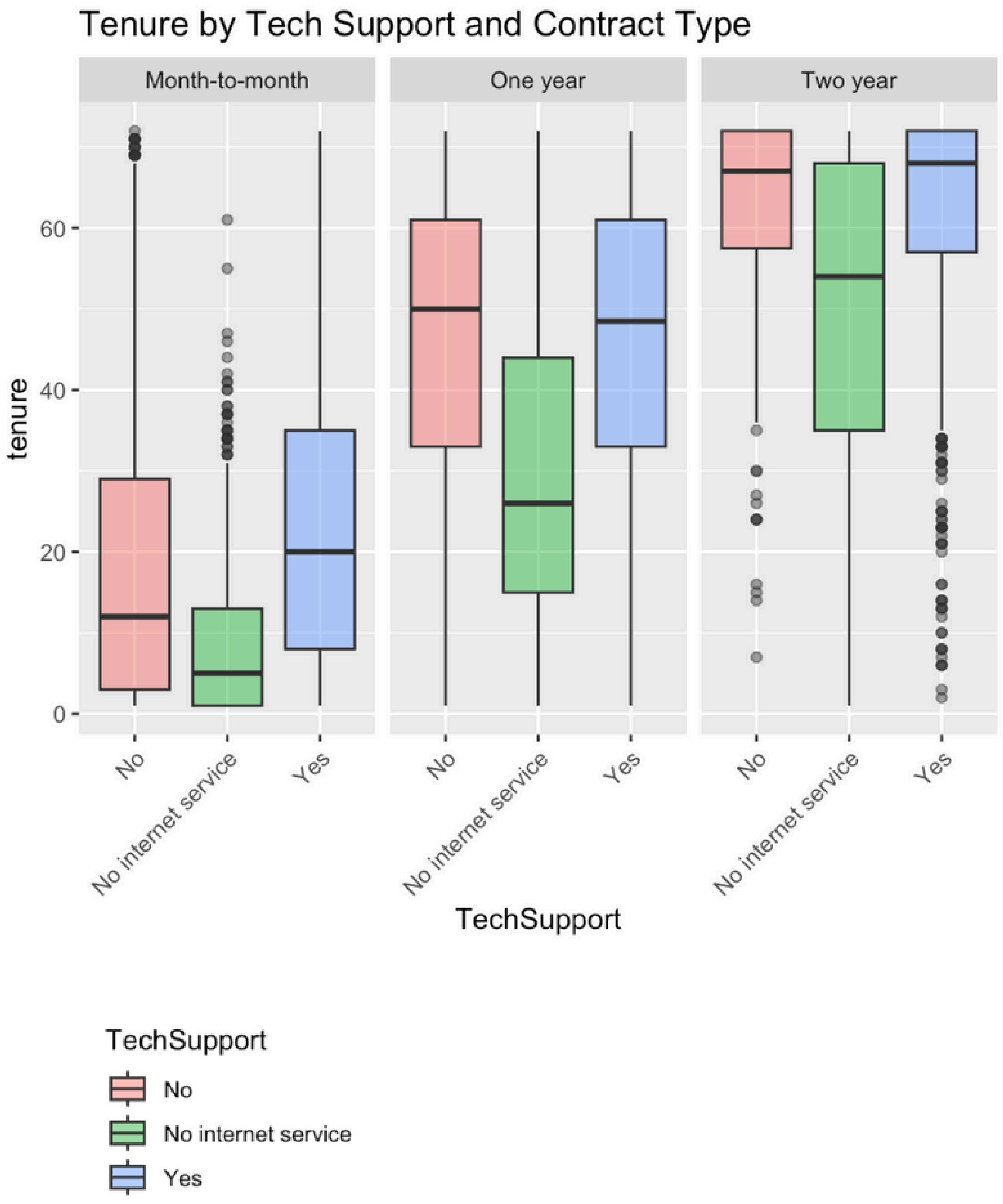




Faceted Plots - by Contract type (Tenure)

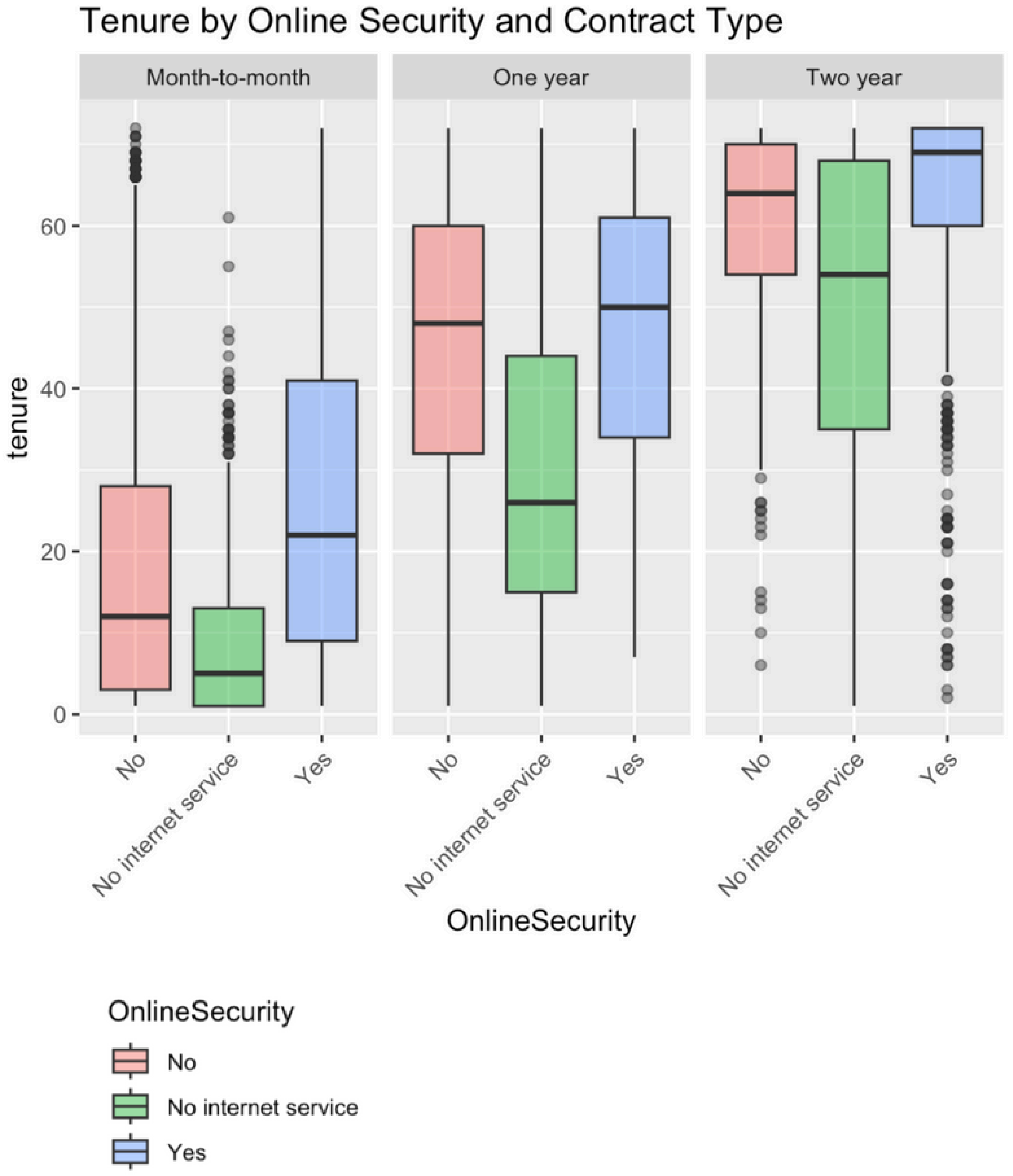
- **Key findings**:
  - Almost identical plots
  - Customers with longer contracts have higher tenure overall.
  - Customers with services consistently show longer tenures compared to others
  - **Outliers -**
    - some customers terminate contract early
    - some month-to-month customers stay long even without longer contracts
  - **Note**: In aggregated box plot services > no internet service > customers
    - But in faceted: services > no services > no internet service.

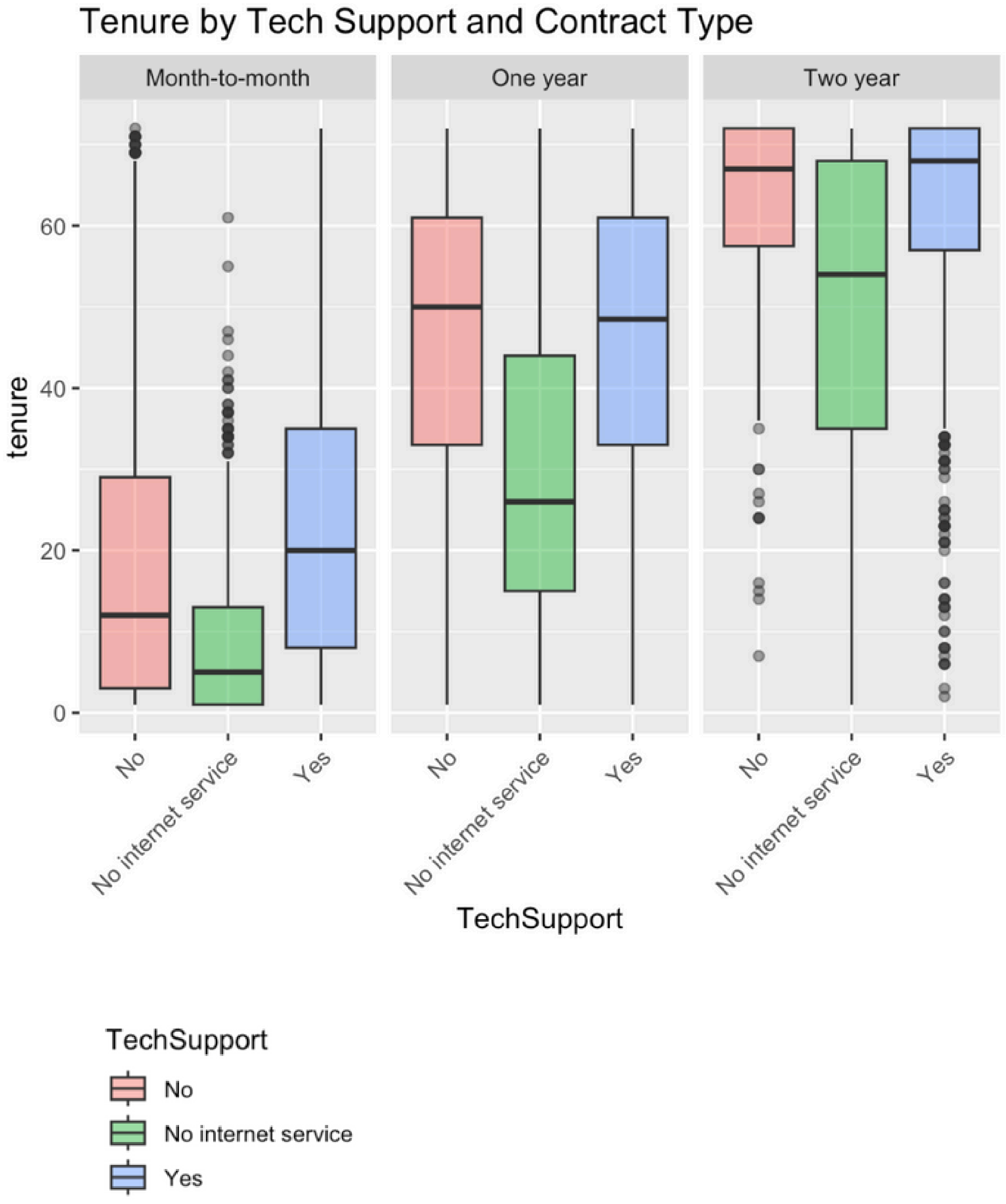




Faceted Plots - by Contract type (total charges)

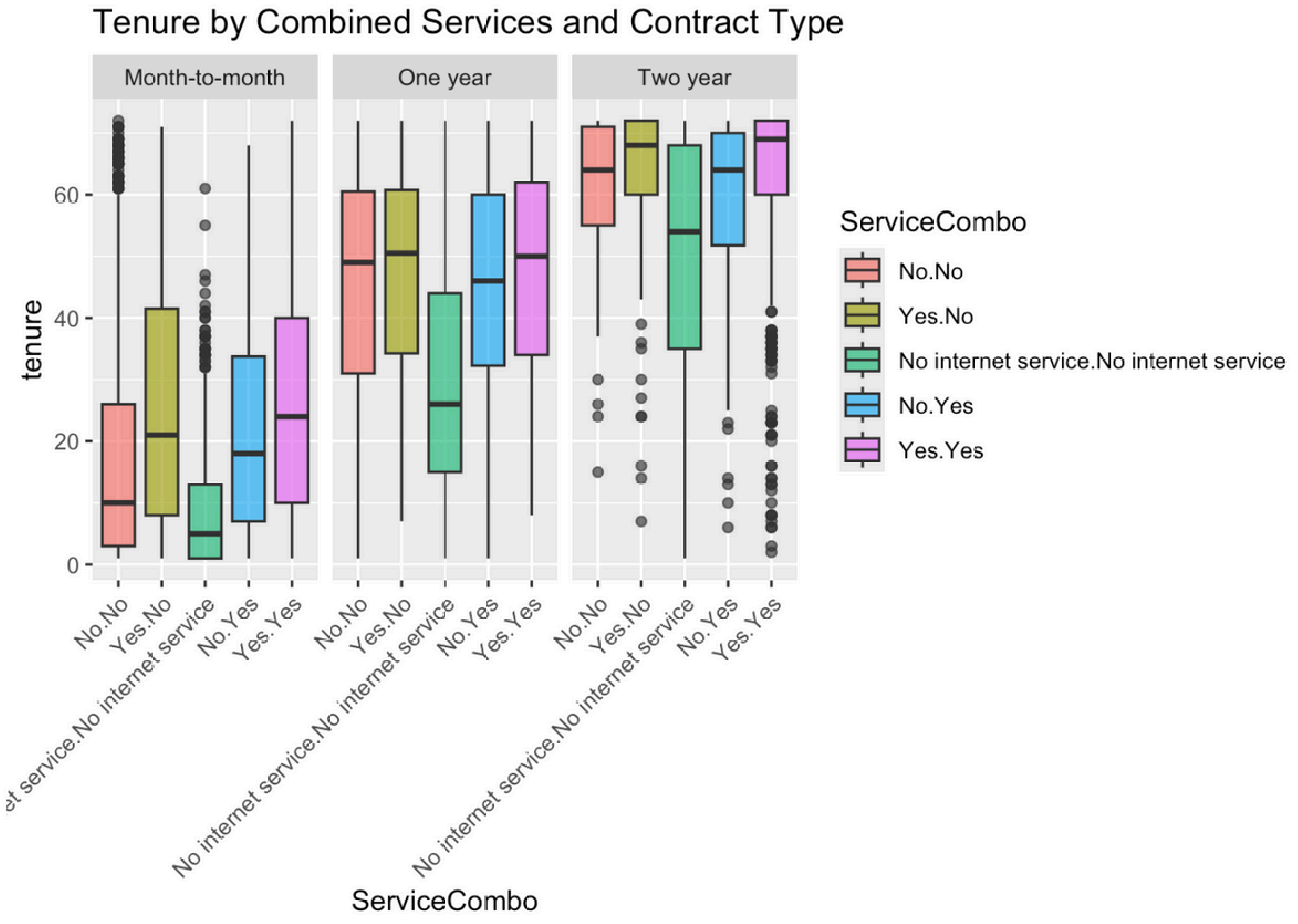
- **Key findings**:
  - Here, the aggregated and faceted plots show same pattern - yes > no > no service
  - Therefore contract type does not confound relationship between service usage and Total Charges.
  - For longer contracts - charges almost same regardless of services
  - Therefore contract length is primary factor
  - For shorter contracts - services have a slightly larger impact on charges





Combined Services and Contract Type Plot

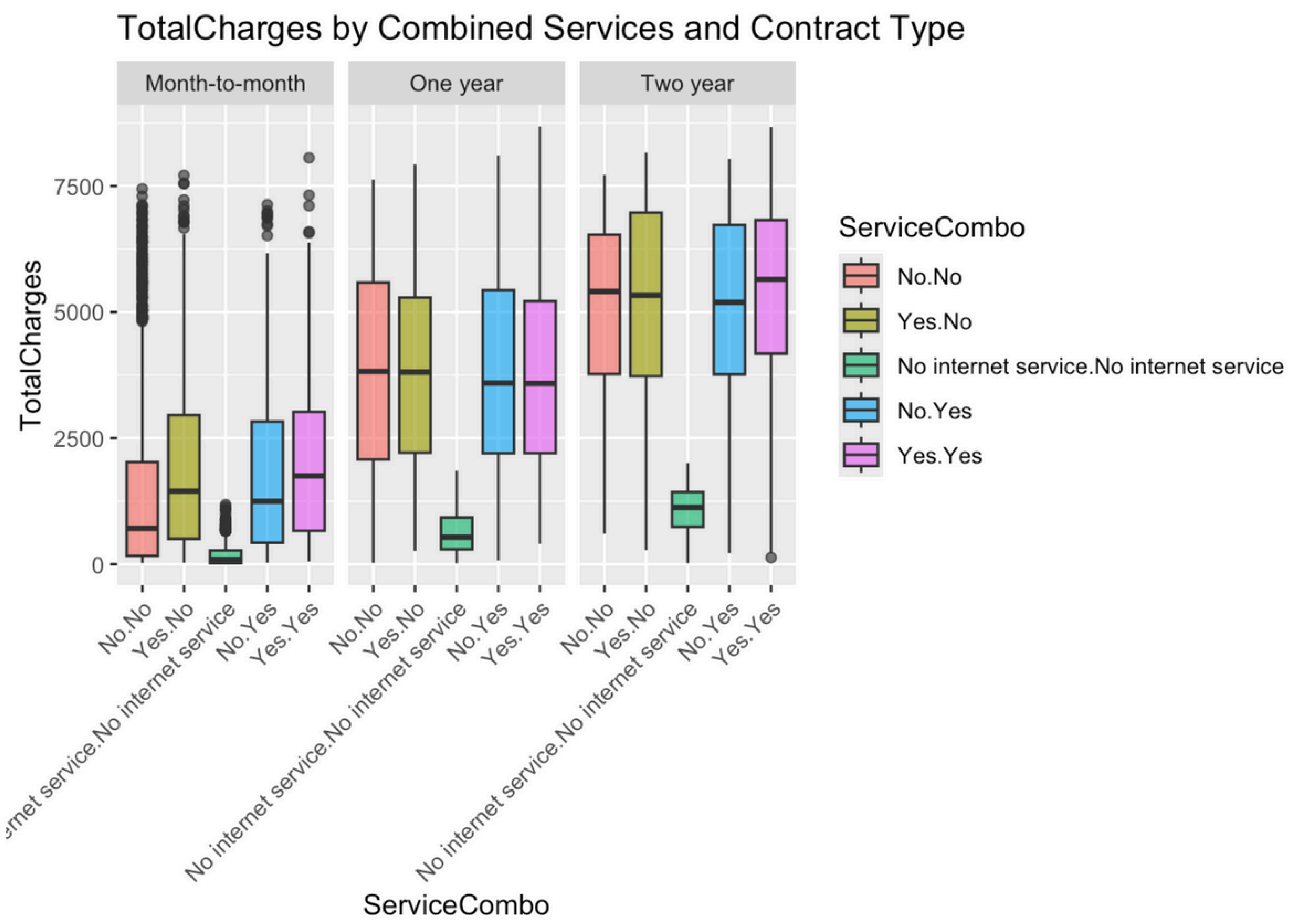
- **Key findings**:
    - The faceted plot shows that contract type is major driver of customer tenure
    - For shorter month-to-month contracts - service usage appears to influence tenure
    - For long-term contracts - customers stay longer regardless of service usage
        - contract commitment outweighs service importance



Combined Services and Contract Type Plot - Total Charges

- **Key findings**:
  - Consistent pattern across contract type indicates Total Charges not confounded by contract type
  - For longer contracts - total charges high regardless of services (cost mostly due to contract length)
  - Shorter contracts - services have slight impact on charges, but less than contract duration

  - Suggests total charges mostly impacted by contract type, and effect of additional charges is small



Conclusion

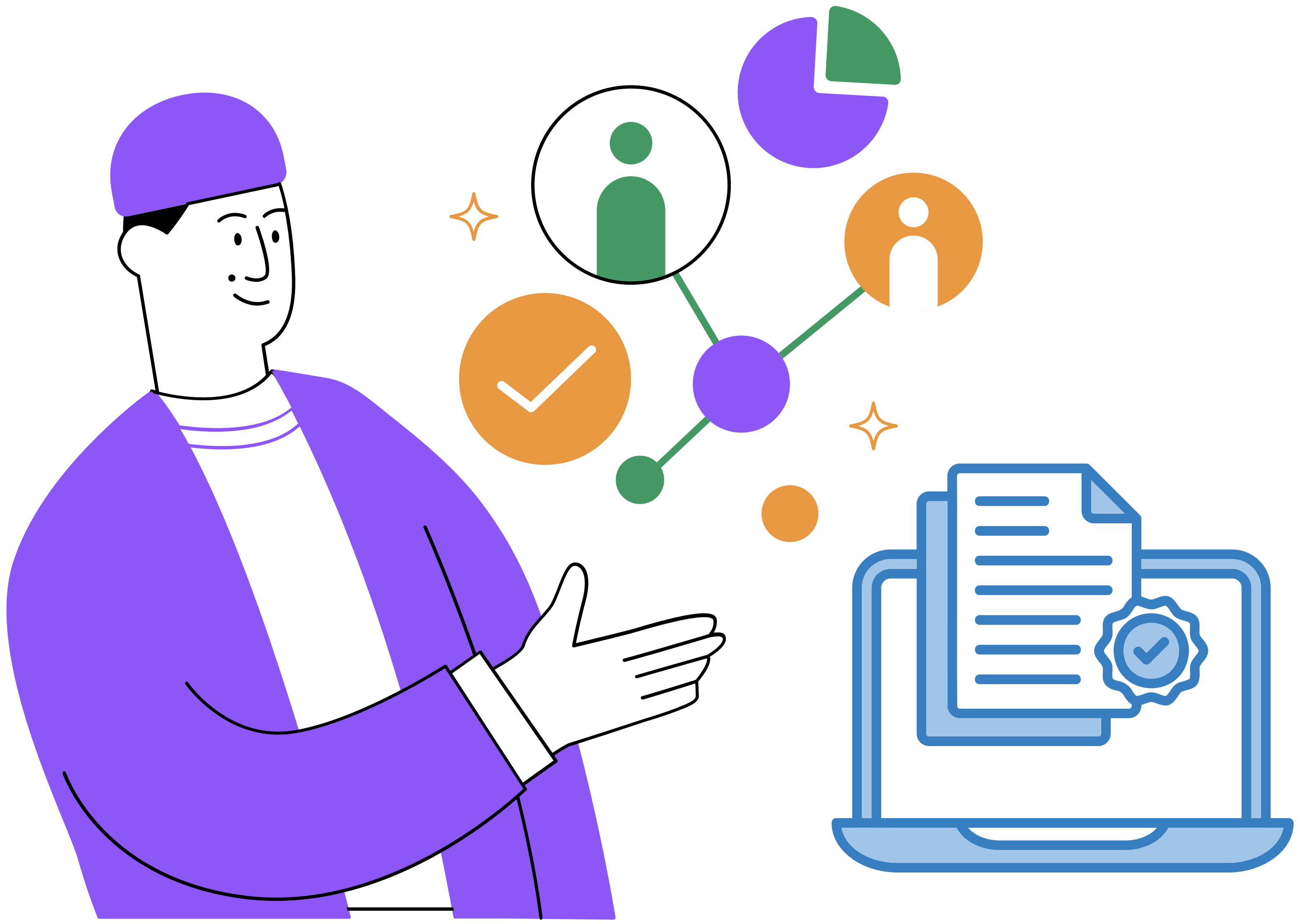
- For month-to-month customers.
  - OnlineSecurity and TechSupport slightly increase tenure and TotalCharges
- For 1-year and 2-year contracts, these services have little to no effect
  - Contract length drives tenure and spending.
- Contract type - most important factor influencing both how long customers stay and how much they spend
- Impact of services - only important for short-term contracts
  - Probably due to customers having more flexibility to leave

Key Business Takeaways

- Targeted Service Bundles:
  - Promote OnlineSecurity and TechSupport to month-to-month customers to boost retention and revenue.
- Upgrade Incentives:
  - Encourage short-term customers to switch to longer contracts using service bundles as value-adds.
- Focus on Contract Design:
  - Since contract length is the strongest retention driver, optimize contract terms and offer personalized contract recommendations based on customer behavior.
- Exit Prevention:
  - For short-term customers without value-added services, implement churn prevention campaigns (e.g., trial TechSupport or OnlineSecurity before contract renewal).

Question 4

What are the primary factors (e.g., combinations of services) that contribute most to high MonthlyCharges, and how do these factors relate to tenure and Churn behavior? Can we identify "premium" customer segments and their churn characteristics?

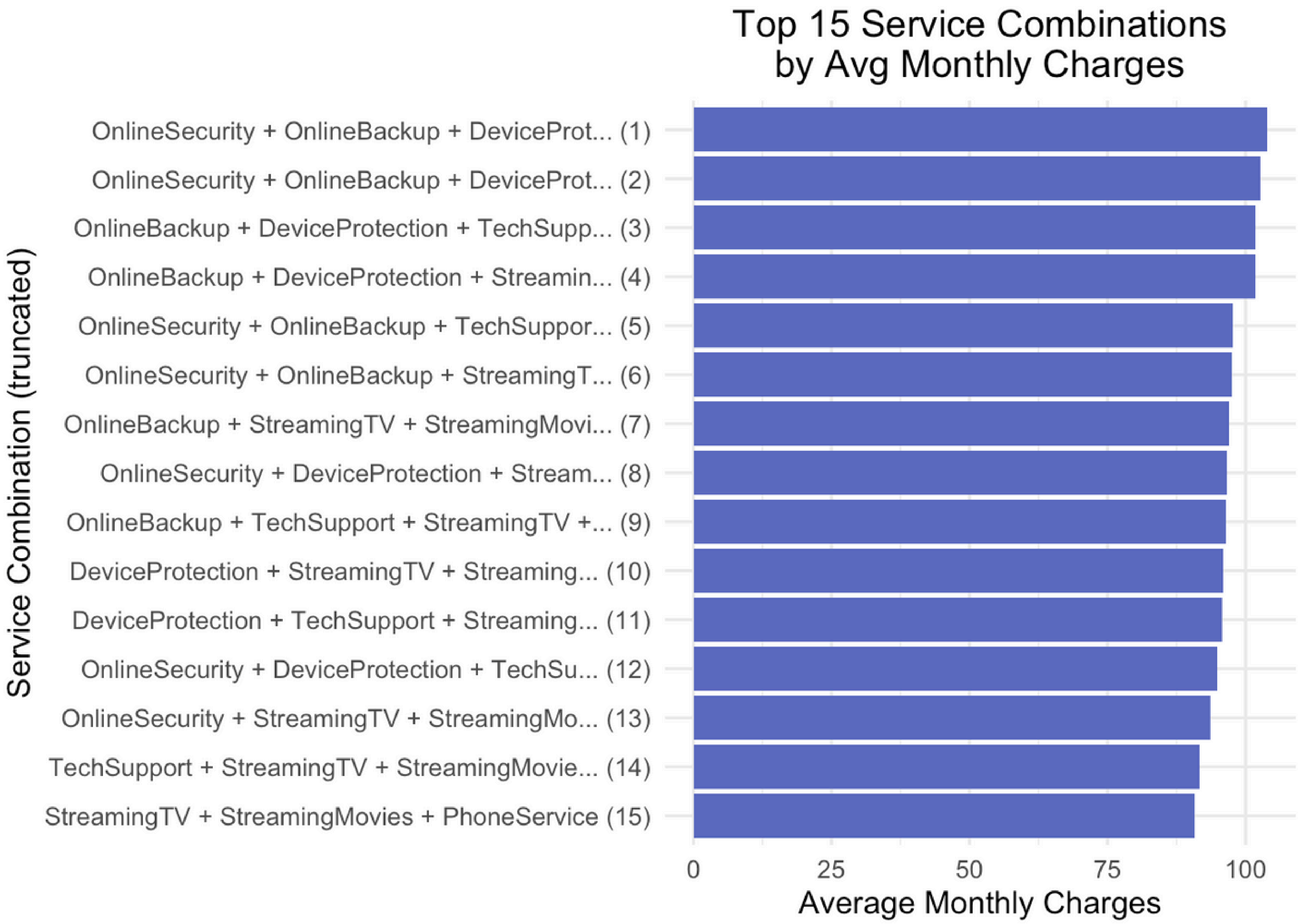


Objective of the Analysis

- Group customers based on the combination of services they use (e.g., phone, internet, security, streaming)
- Identify which service combos lead to **high MonthlyCharges**
- Analyze how these combos **relate to tenure and churn**
- Define potential **premium segments** (high-paying, multi-service users)
- Understand churn risk and retention priorities for each segment

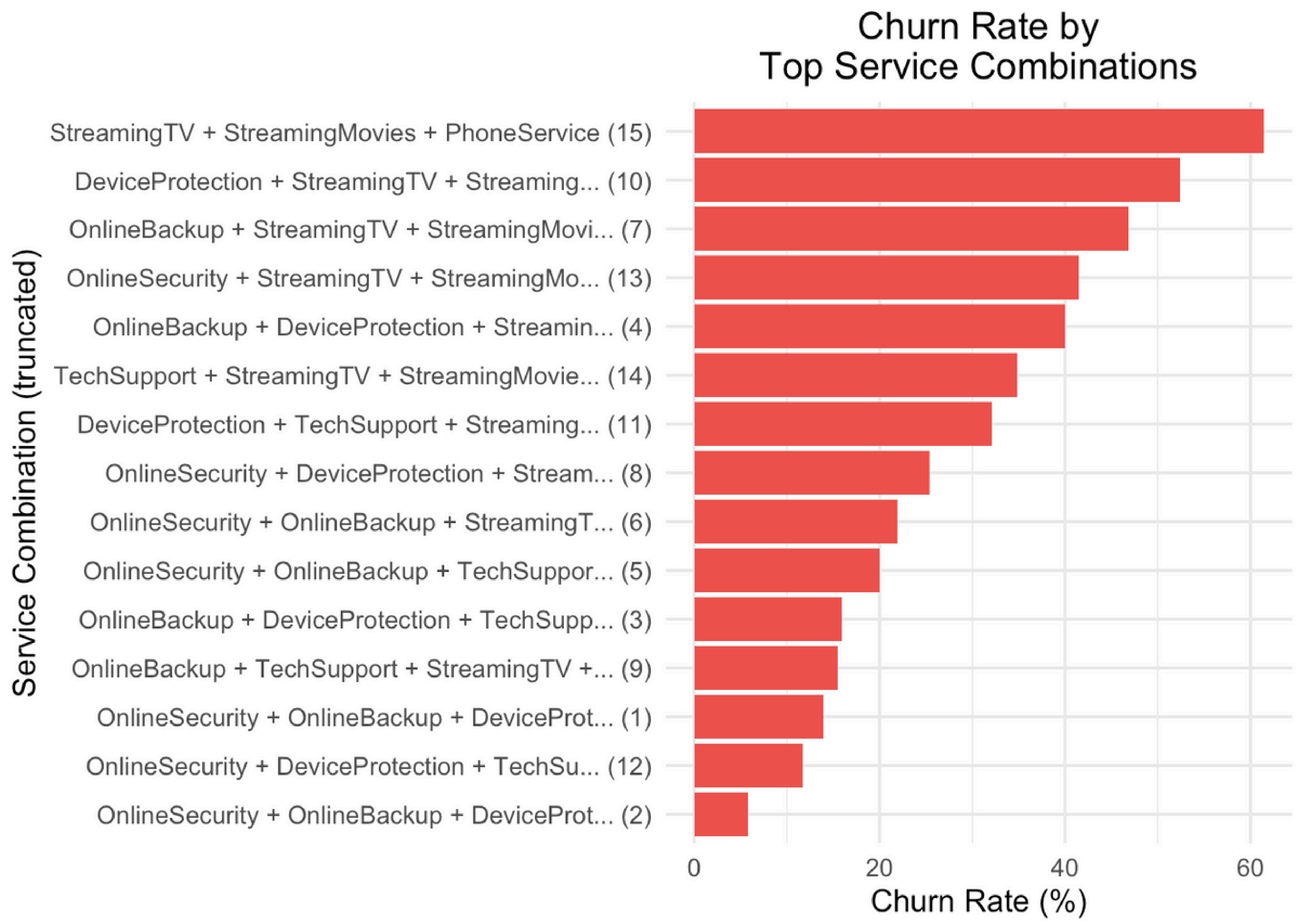
Average Monthly Charges by Service Combination

- **Key findings**:
  - Service combinations with 5+ add-ons lead to the highest monthly charges.
  - Combos involving StreamingTV, OnlineSecurity, and TechSupport dominate top ranks.
  - Customers subscribing to these bundled services are likely to be heavy users.



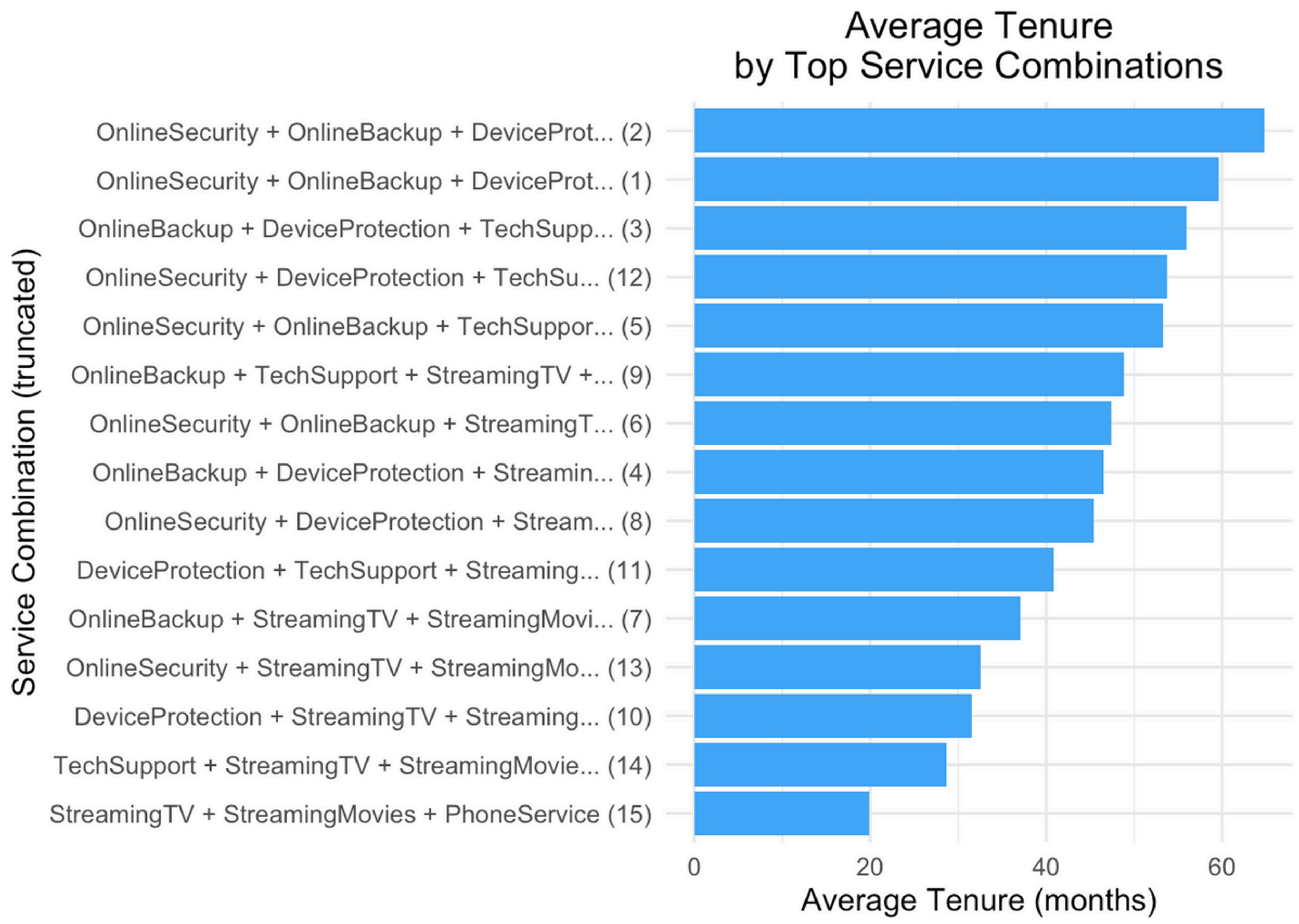
Churn Rate for Top Combinations

- **Key findings**:
  - Some high-charge combos also churn more frequently.
  - Heavy churn in combos with streaming but no support services.
  - Lower churn in combos that include TechSupport and OnlineSecurity.



Average Tenure by Service Combination

- **Key findings**:
  - Many high-charge customers have short tenure → indicating they churn early.
  - Some combinations show both high charges and long tenure, likely loyal premium users.
  - Significant variation in customer lifetime value across combos.



What are Premium Segments for us?

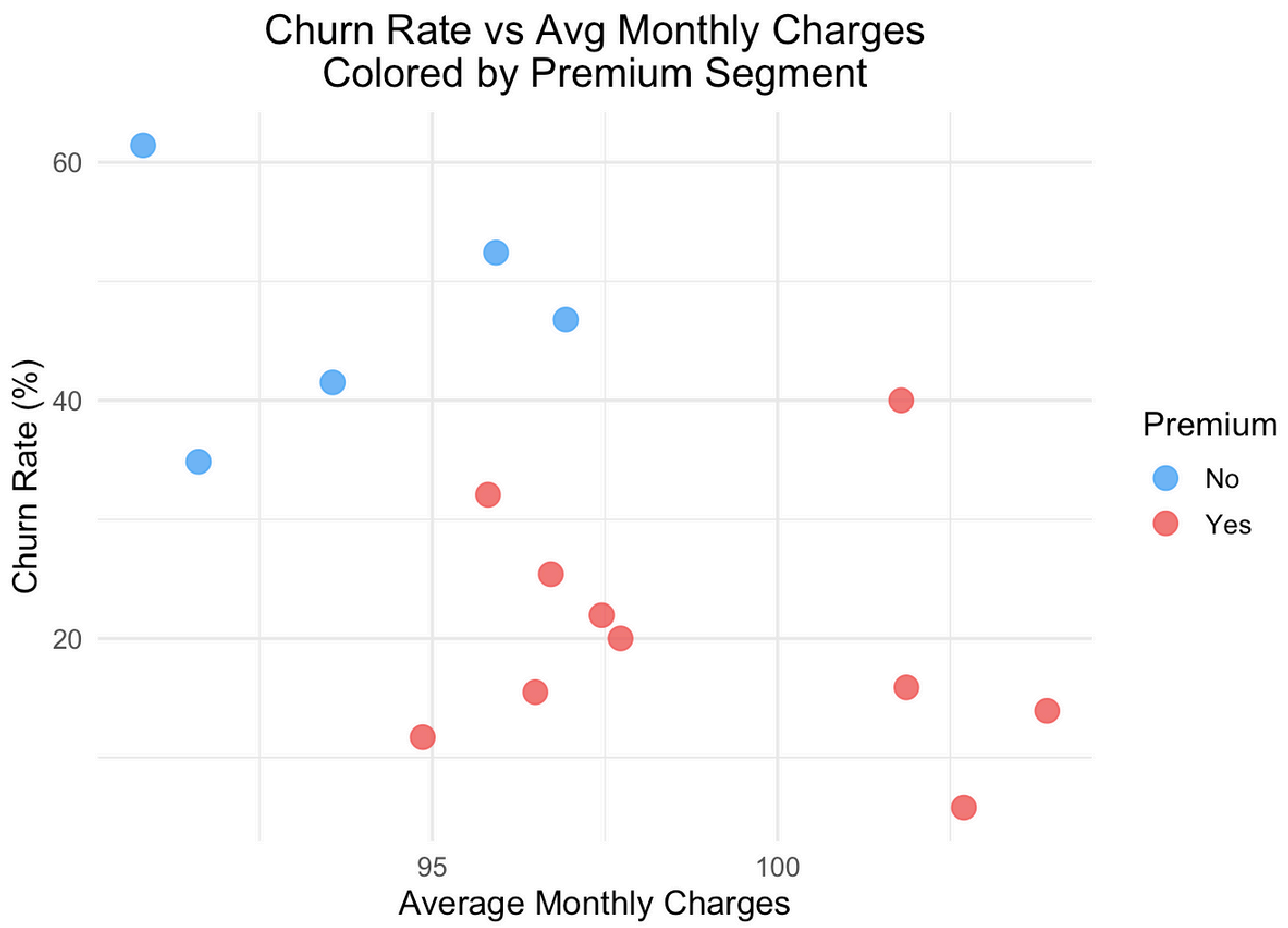
- Premium defined as **avg charge > $90** and **5+ services**.
- These segments have **high revenue potential** but also **shorter tenure** and **higher churn**.
- Their service usage and billing place them among the highest-value customers.

Identify Premium Customer Segments

- **Key findings**:
  - **Premium combos (red)** mostly cluster in the top-right:
  - High MonthlyCharges
  - High ChurnRate

  - **Non-premium segments (blue)** are concentrated lower-left:
  - Low cost, low churn — stable and low-risk

  - This clearly shows that **high revenue = high risk,** and targeted retention is needed for premium users.

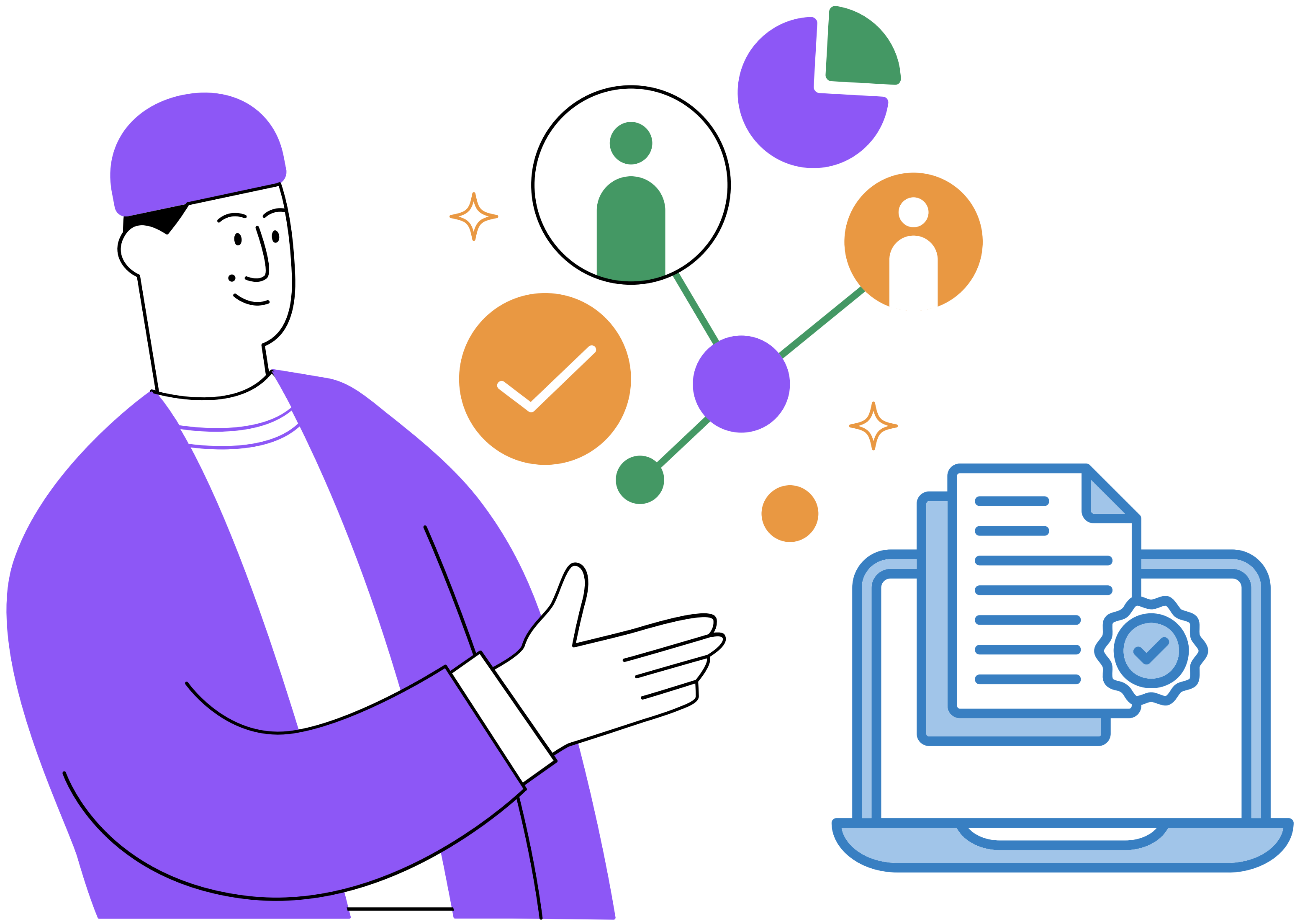


Key Business Takeaways

- Reward premium users (5+ services) with exclusive perks (e.g., discounts, upgrades, VIP support) to reduce early churn and enhance satisfaction.
- Combine OnlineSecurity and TechSupport with high-risk service bundles (like streaming-heavy plans) to stabilize churn without reducing price.
- Redirect some marketing spend to retention initiatives (e.g., churn prediction, win-back campaigns) for premium segments instead of always acquiring new customers.
- Targeted retention (e.g., loyalty perks, better onboarding) is essential for sustaining premium users.
- Customer lifetime value varies widely — segmentation helps identify where to focus efforts.

Question 5

Can we identify distinct customer segments based on service usage patterns, and how do these segments differ in churn, loyalty, and spending?

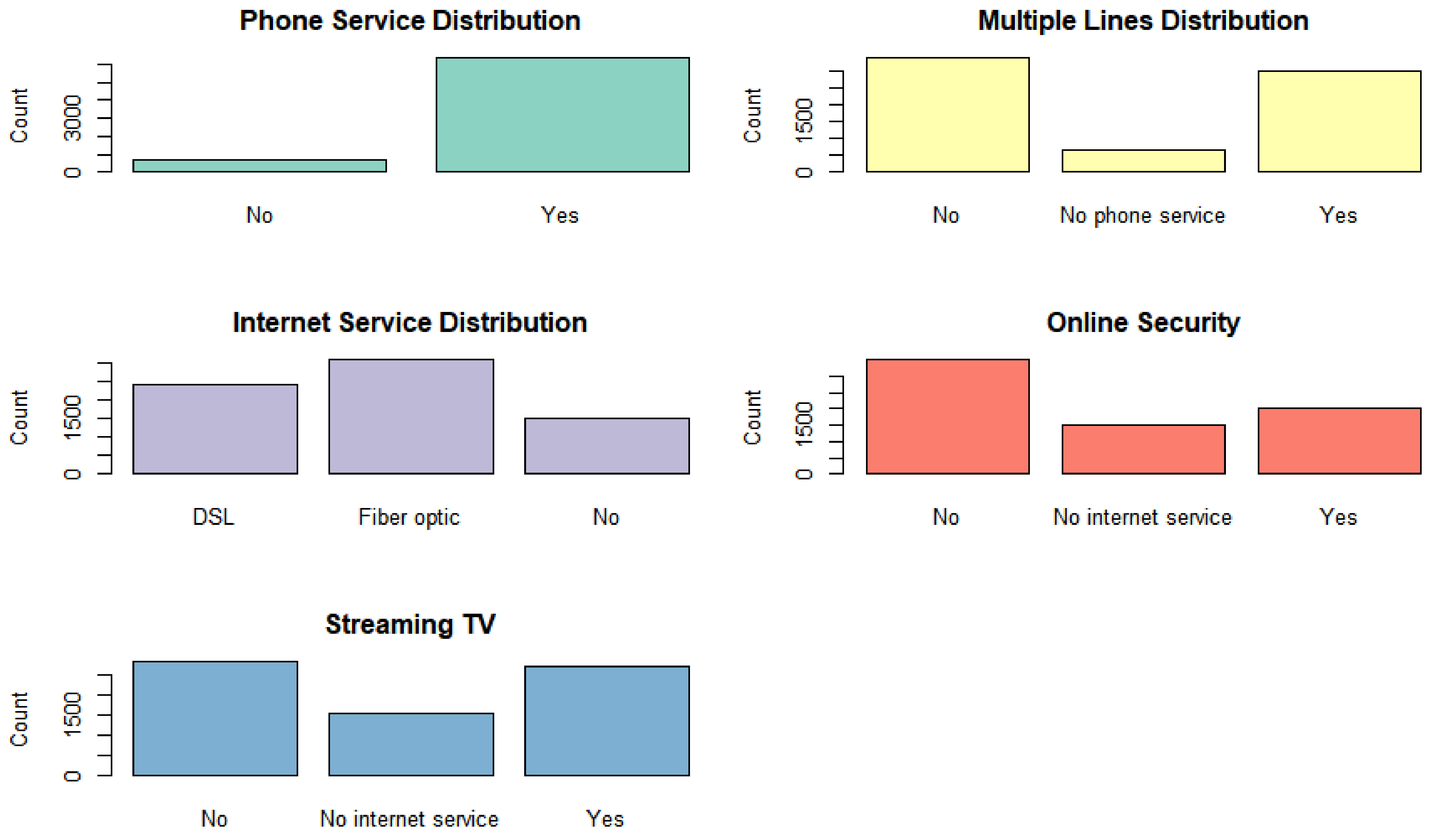


Objective of the Analysis

- Group customers into meaningful segments based on all the services they use such as phone, internet, security, streaming, etc.
- Once we've grouped them (using clustering), we want to compare those groups to see:
- Churn (Who is more likely to leave?)
- Monthly charges (Who spends more?)
- Tenure (How long have they stayed?)

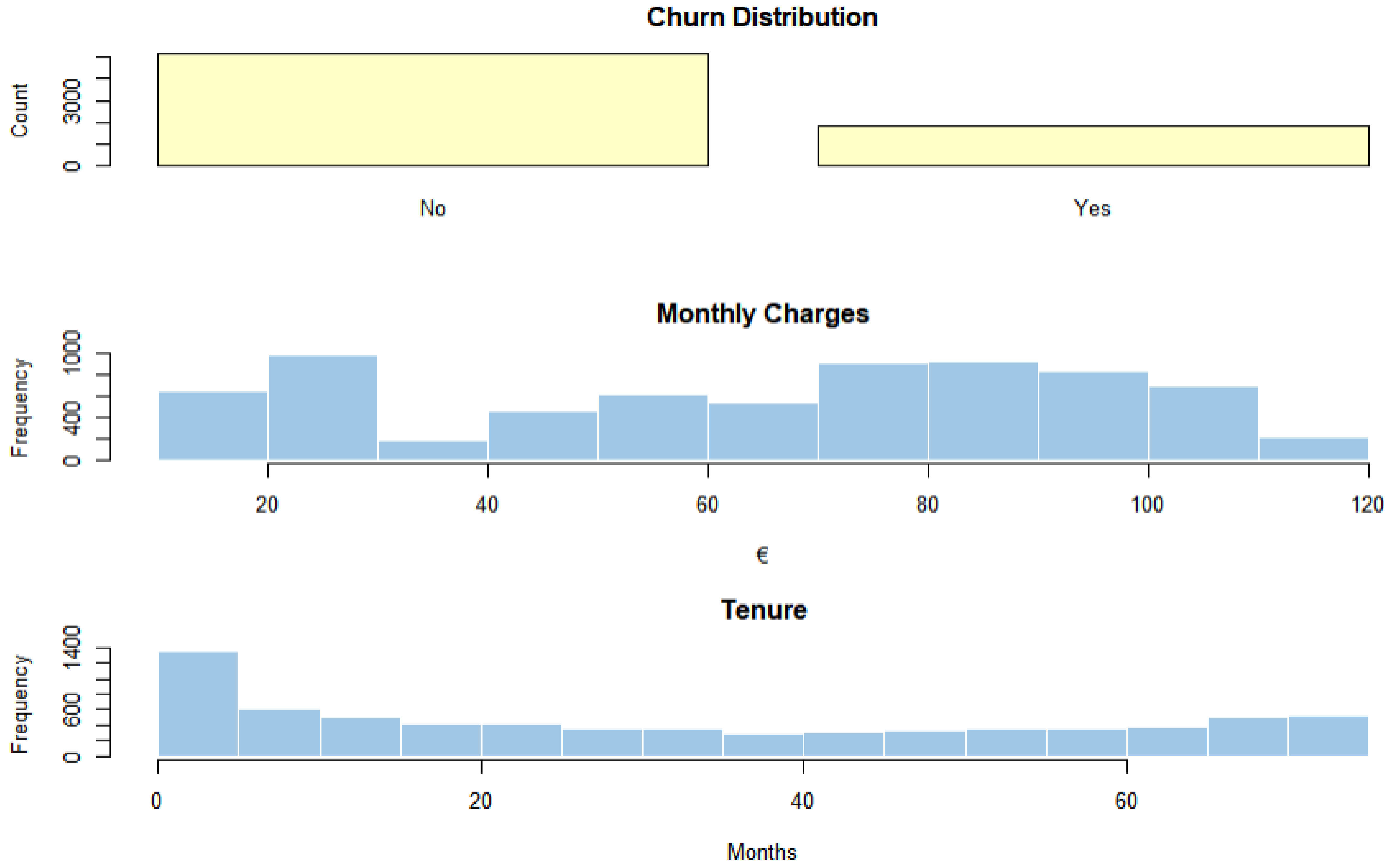
Service Usage Patterns

- **Phone Service Distribution**: Most customers have phone service.
- **Multiple Lines Distribution**: Many have multiple lines, but a significant portion do not.
- **Internet Service Distribution**: Most use DSL or Fiber optic; fewer have no internet.
- **Online Security** & **Streaming TV**: Many customers do not have these optional add-ons, though some do.



Churn, Spending & Loyalty

- **Churn:** Most customers stayed, but a notable number have churned.
- **Monthly Charges:** Most customers pay €20–€80, peaking at €20–30; few pay above €80.
- **Tenure:** Most customers have short tenures (0–10 months), and the frequency gradually decreases, with fewer customers having longer tenures (40+ months).



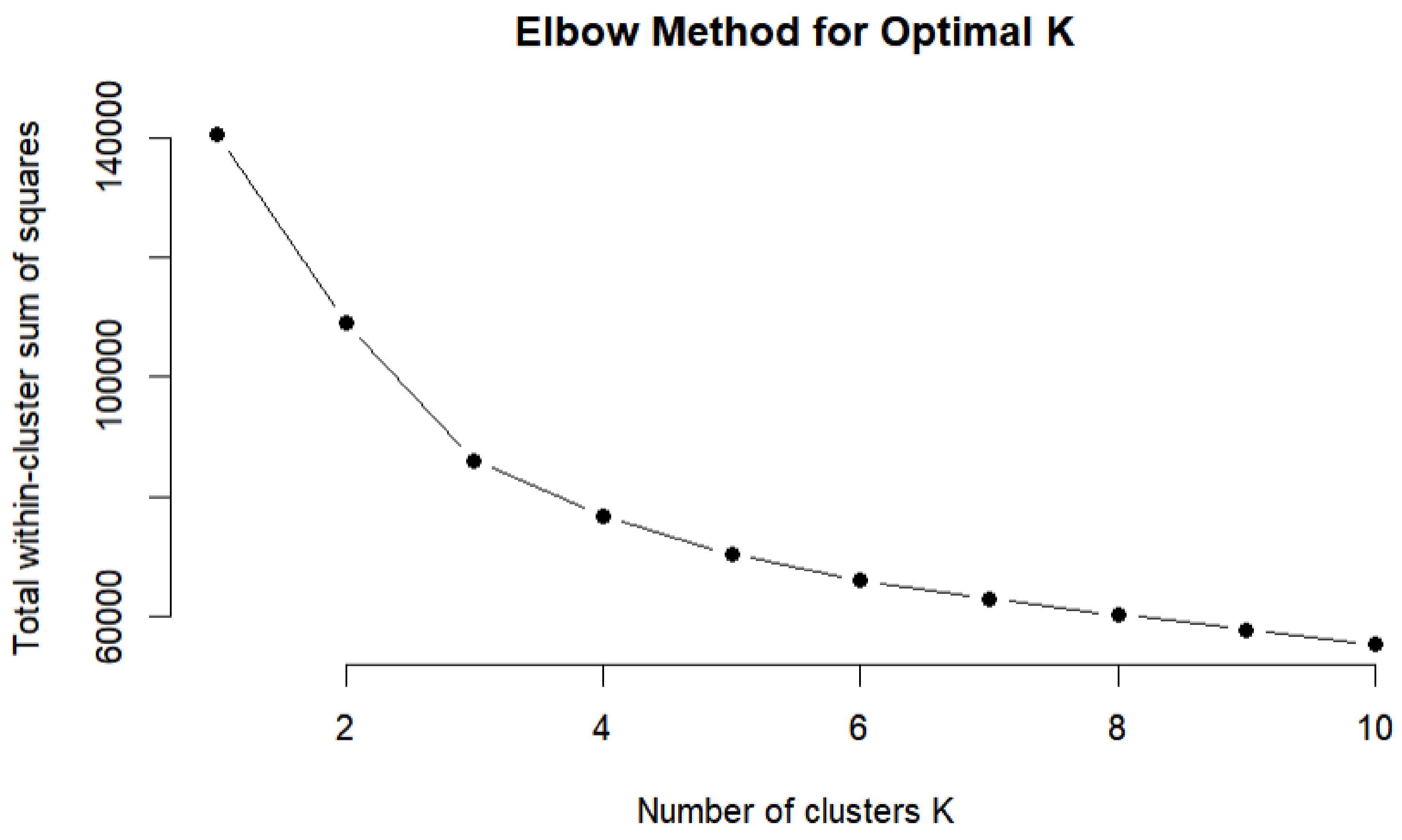
Preparing for Clustering & Choosing Number of Groups

Preparing Data for Clustering

- Applied one-hot encoding to turn each option into a separate binary feature.
- Finally, scaled the encoded values so no service dominates the clustering due to higher frequency.

Finding Optimal Number of Clusters (K)

- Used the Elbow Method by running K-means for K = 1 to 10.
- Measured how tightly customers group in each K using WSS (Within-Cluster Sum of Squares).



Applying K-Means Clustering

Clustering Process

- We used K-means clustering to segment customers based on their service usage patterns.
- Based on the Elbow Method, we chose K = 4 as the ideal number of customer segments.
- To ensure stability, the algorithm was initialized 25 times (nstart = 25), reducing the risk of random, suboptimal clusters.
- Each customer was assigned to a Cluster (1 to 4) — this new variable helps us analyze differences in churn, spending, and loyalty.

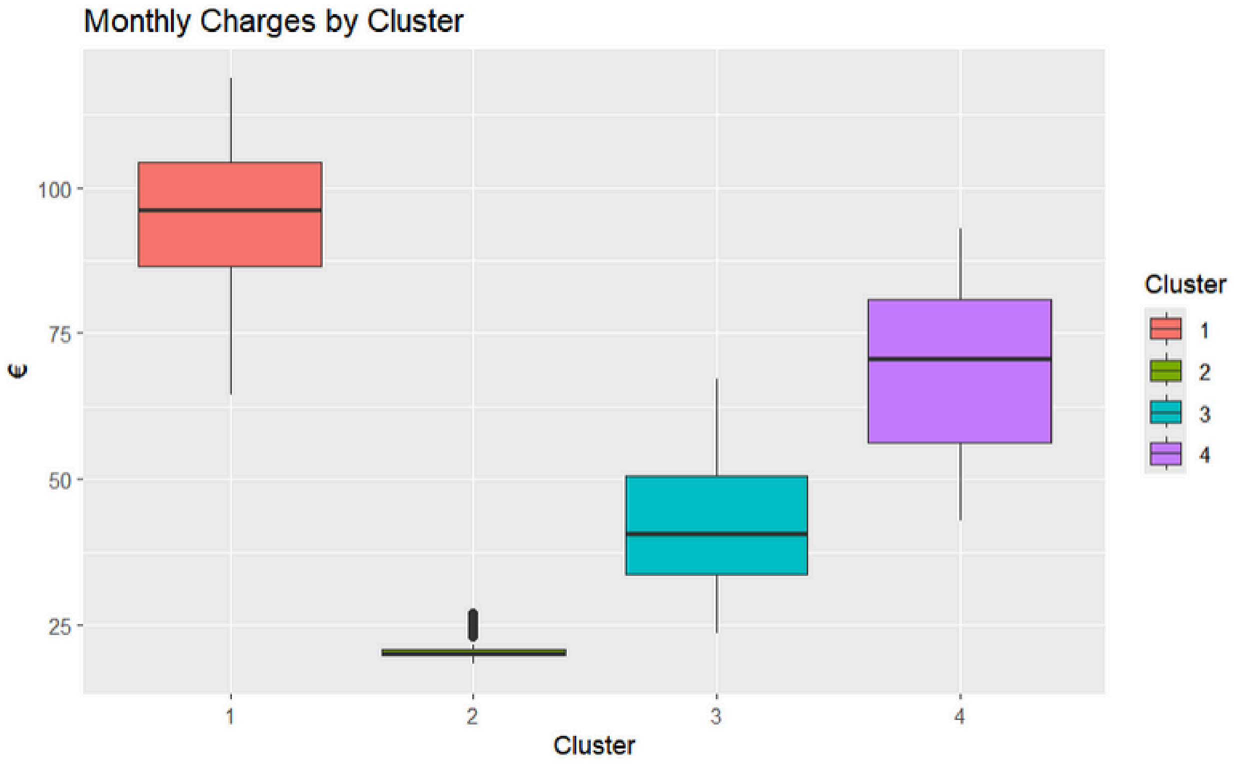

```
```{r}
# Apply K-means clustering (k = 4 from above elbow method)
set.seed(123)
k <- 4
kmeans_result <- kmeans(cluster_scaled, centers = k, nstart = 25)
data$Cluster <- as.factor(kmeans_result$cluster)

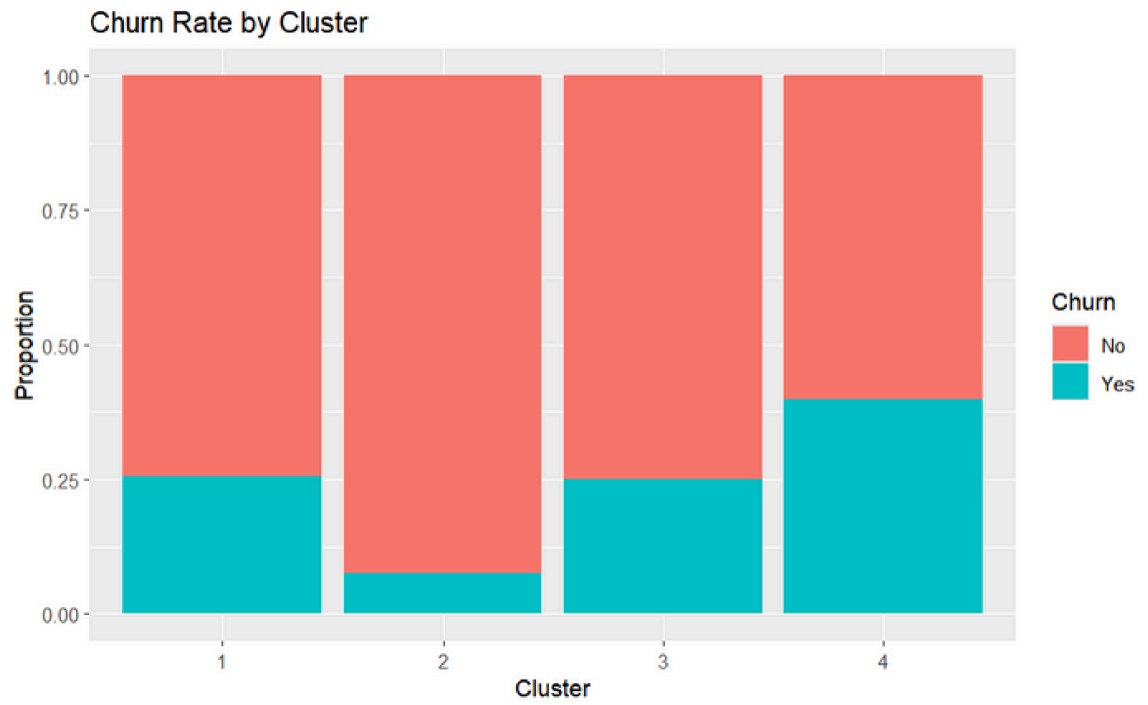
```
```

Comparing Clusters by Churn, Spending, and Loyalty

Monthly Charges

- Cluster 1: Highest spenders → Use many services.
- Cluster 2: Lowest spenders → Basic users.
- Cluster 3: Mid-range → Potential upsell group.
- Cluster 4: Mixed spenders → Diverse needs.



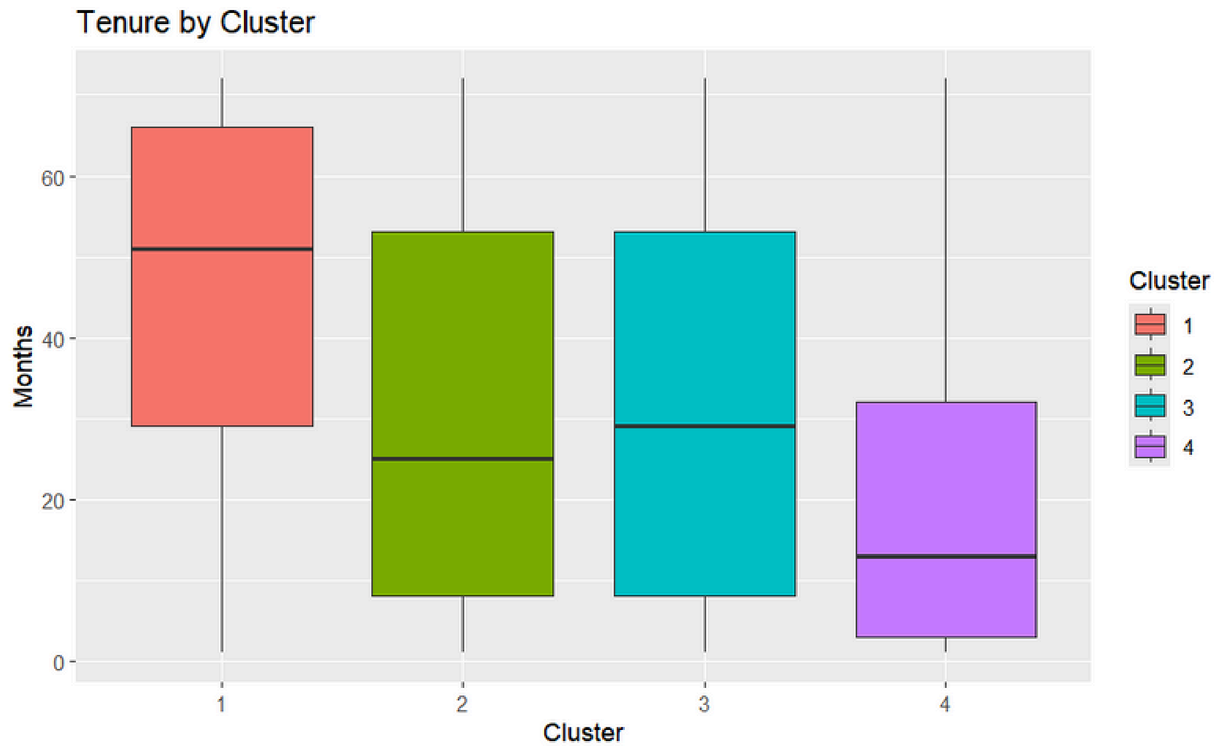


Churn Distribution

- Cluster 2: Lowest churn → Very loyal.
- Cluster 4: Highest churn → At-risk customers.
- Clusters 1 & 3: Moderate churn (~25%).

Tenure

- Cluster 1: Longest tenure → Loyal & long-term.
- Cluster 2: Mixed → Old + new.
- Cluster 3: Mid-range tenure.
- Cluster 4: Shortest → New customers.



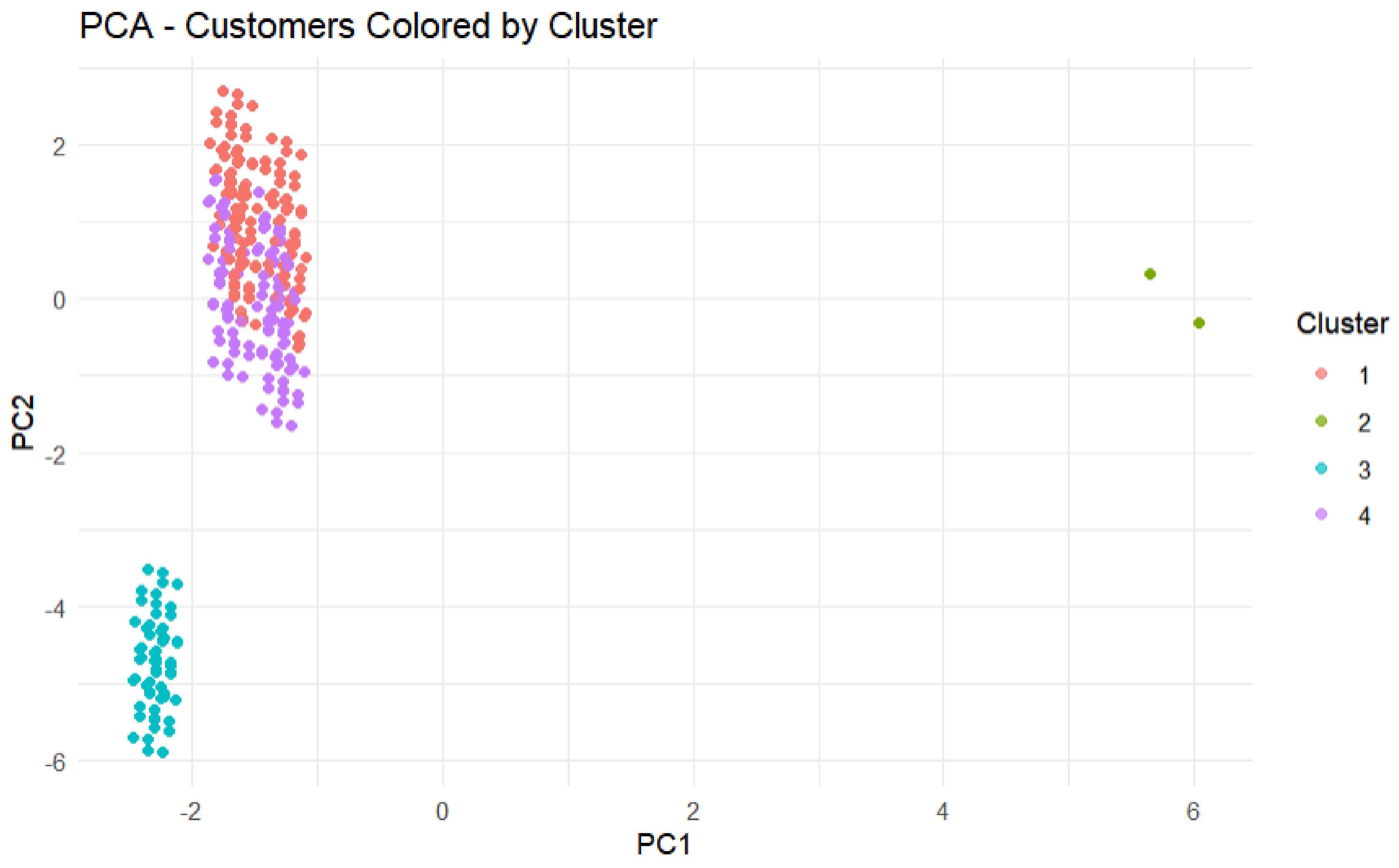
Visualizing Customer Segments via PCA

Principal Component Analysis (PCA)

- PCA helps reduce dimensions of the data for better visualization.
- PC1 captures 37.9%, and PC2 adds 14.3% → Over 52% of the data's structure is preserved in just 2D.

PCA Insights

- **Cluster 2** is well-separated → likely basic, low-engagement users.
- **Cluster 1 & 4** slightly overlap, but differ in tenure/spending.
- **Cluster 3** forms a distinct group → moderate users with consistent behavior.



Key Business Takeaways

- In **Cluster 4**: These customers leave the most. Try to understand why and offer better support or deals to keep them.
- In **Cluster 3**: They use some services but might want more. Suggest useful add-ons or upgrades.
- In **Cluster 1**: They pay a lot and stay long. Give them perks or discounts to keep them happy.
- In **Cluster 2**: They are low-paying and loyal, but if something changes, they might leave. Keep it simple and affordable for them.

Cluster	Churn Rate	Avg Monthly Charges	Avg Tenure
1	~25%	High (€95)	Long
2	**~7%**	**Low (€21)**	Moderate
3	~25%	Moderate (€42)	Moderate
4	**~40%**	Medium (€69)	**Low**

Thank You